Code

Loan Default Risk Analysis and Prediction Using R

Case study: predicting loan default with Logistic Regression and Random Forest

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> **Reproducibility note:** Place `credit_risk_dataset.csv` in the same folder as this `.Rmd` file before knitting. The data path is relative, so the report runs on any computer. All tables, numbers and plots are regenerated when the report is knitted; they are not copied from the notebook. Small differences from the originally reported figures are possible if package versions differ.

1 Case Study Overview

1.1 Problem statement

Loan default creates financial risk for lenders. This case study explores borrower and loan characteristics associated with default and builds predictive models in R to estimate whether a loan is likely to default.

1.2 Objectives

- Clean and prepare the credit-risk dataset.
- Explore default patterns through tables and visualizations.
- Examine numerical correlations and categorical associations with loan status (chi-square tests).
- Train and evaluate Logistic Regression and Random Forest models on a stratified 80:20 train-test split.
- Compare the models using accuracy, precision, recall, F1-score, ROC curves and AUC.
- Examine Random Forest feature importance.
- Run a probability-threshold experiment and a class-weighted Random Forest experiment.
- Use LIME to inspect one individual Random Forest prediction.

2 Dataset

Source: Credit Risk Dataset on Kaggle - https://www.kaggle.com/datasets/laotse/credit-risk-dataset (https://www.kaggle.com/datasets/laotse/credit-risk-dataset). The local file is `credit_risk_dataset.csv` .

The original file has 32,581 rows and 12 variables. The target is `loan_status` , where `0` denotes non-default and `1` denotes default.

Variable	Description
`person_age`	Age of the borrower (years)
`person_income`	Annual income of the borrower

Variable	Description
person_home_ownership	Home ownership status (RENT, OWN, MORTGAGE, OTHER)
person_emp_length	Employment length (years)
loan_intent	Purpose of the loan
loan_grade	Loan grade (A to G)
loan_amnt	Loan amount
loan_int_rate	Loan interest rate (%)
loan_status	Target: 0 = non-default, 1 = default
loan_percent_income	Loan amount as a fraction of income
cb_person_default_on_file	Historical default on file (Y/N)
cb_person_cred_hist_length	Credit history length (years)

3 Setup and Data Import

3.1 Load required packages

All packages are loaded once, here, so every later chunk can rely on them.

Show

```
## All packages loaded successfully.
```

3.2 Import the dataset

Show

```
## Dataset imported successfully.
```

Show

```
## Rows: 32581
```

Show

```
## Columns: 12
```

Show

```
##   person_age person_income person_home_ownership person_emp_length loan_intent
## 1         22         59000                  RENT               123    PERSONAL
## 2         21          9600                   OWN                 5   EDUCATION
## 3         25          9600              MORTGAGE                 1     MEDICAL
## 4         23         65500                  RENT                 4     MEDICAL
## 5         24         54400                  RENT                 8     MEDICAL
## 6         21          9900                   OWN                 2     VENTURE
##   loan_grade loan_amnt loan_int_rate loan_status loan_percent_income
## 1          D     35000         16.02           1                0.59
## 2          B      1000         11.14           0                0.10
## 3          C      5500         12.87           1                0.57
## 4          C     35000         15.23           1                0.53
## 5          C     35000         14.27           1                0.55
## 6          A      2500          7.14           1                0.25
##   cb_person_default_on_file cb_person_cred_hist_length
## 1                         Y                          3
## 2                         N                          2
## 3                         N                          3
## 4                         N                          2
## 5                         Y                          4
## 6                         N                          2
```

4 Data Inspection and Cleaning

4.1 Initial inspection and summary statistics

Show

```
## [1] 32581    12
```

Show

```
## 'data.frame':    32581 obs. of  12 variables:
##  $ person_age                : int  22 21 25 23 24 21 26 24 24 21 ...
##  $ person_income             : int  59000 9600 9600 65500 54400 9900 77100 78956 83000 100
00 ...
##  $ person_home_ownership     : chr  "RENT" "OWN" "MORTGAGE" "RENT" ...
##  $ person_emp_length         : num  123 5 1 4 8 2 8 5 8 6 ...
##  $ loan_intent               : chr  "PERSONAL" "EDUCATION" "MEDICAL" "MEDICAL" ...
##  $ loan_grade                : chr  "D" "B" "C" "C" ...
##  $ loan_amnt                 : int  35000 1000 5500 35000 35000 2500 35000 35000 35000 160
0 ...
##  $ loan_int_rate             : num  16 11.1 12.9 15.2 14.3 ...
##  $ loan_status               : int  1 0 1 1 1 1 1 1 1 1 ...
##  $ loan_percent_income       : num  0.59 0.1 0.57 0.53 0.55 0.25 0.45 0.44 0.42 0.16 ...
##  $ cb_person_default_on_file : chr  "Y" "N" "N" "N" ...
##  $ cb_person_cred_hist_length: int  3 2 3 2 4 2 3 4 2 3 ...
```

Show

```
##    person_age     person_income     person_home_ownership person_emp_length
##  Min.   : 20.00   Min.   :   4000   Length   :32581       Min.   :  0.00
##  1st Qu.: 23.00   1st Qu.:  38500   N.unique :    4       1st Qu.:  2.00
##  Median : 26.00   Median :  55000   N.blank  :    0       Median :  4.00
##  Mean   : 27.73   Mean   :  66075   Min.nchar:    3       Mean   :  4.79
##  3rd Qu.: 30.00   3rd Qu.:  79200   Max.nchar:    8       3rd Qu.:  7.00
##  Max.   :144.00   Max.   :6000000                         Max.   :123.00
##                                                           NAs    :895
##     loan_intent        loan_grade      loan_amnt     loan_int_rate
##  Length   :32581   Length   :32581   Min.   :  500   Min.   : 5.42
##  N.unique :    6   N.unique :    7   1st Qu.: 5000   1st Qu.: 7.90
##  N.blank  :    0   N.blank  :    0   Median : 8000   Median :10.99
##  Min.nchar:    7   Min.nchar:    1   Mean   : 9589   Mean   :11.01
##  Max.nchar:   17   Max.nchar:    1   3rd Qu.:12200   3rd Qu.:13.47
##                                      Max.   :35000   Max.   :23.22
##                                                      NAs    :3116
##   loan_status     loan_percent_income cb_person_default_on_file
##  Min.   :0.0000   Min.   :0.0000      Length   :32581
##  1st Qu.:0.0000   1st Qu.:0.0900      N.unique :    2
##  Median :0.0000   Median :0.1500      N.blank  :    0
##  Mean   :0.2182   Mean   :0.1702      Min.nchar:    1
##  3rd Qu.:0.0000   3rd Qu.:0.2300      Max.nchar:    1
##  Max.   :1.0000   Max.   :0.8300
##
##  cb_person_cred_hist_length
##  Min.   : 2.000
##  1st Qu.: 3.000
##  Median : 4.000
##  Mean   : 5.804
##  3rd Qu.: 8.000
##  Max.   :30.000
##
```

Show

```
##                person_age             person_income
##                         0                         0
##      person_home_ownership          person_emp_length
##                         0                       895
##               loan_intent               loan_grade
##                         0                         0
##                 loan_amnt             loan_int_rate
##                         0                      3116
##               loan_status        loan_percent_income
##                         0                         0
##  cb_person_default_on_file cb_person_cred_hist_length
##                         0                         0
```

Show

```
##
##     0     1
## 25473  7108
```

4.2 Data quality checks: duplicates, outliers and unusual values

Show

```
## [1] 165
```

Show

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   20.00   23.00   26.00   27.73   30.00  144.00
```

Show

```
## [1] 7
```

Show

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
##    0.00    2.00    4.00    4.79    7.00  123.00     895
```

Show

```
## [1] 2
```

Show

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    4000   38500   55000   66075   79200 6000000
```

Show

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     500    5000    8000    9589   12200   35000
```

Show

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
##    5.42    7.90   10.99   11.01   13.47   23.22    3116
```

4.3 Inspect unusual records

Show

```
##       person_age person_emp_length person_income
## 82           144                 4        250000
## 184          144                 4        200000
## 576          123                 2         80004
## 748          123                 7         78000
## 32298        144                12       6000000
## 32417         94                 1         24000
## 32507         84                 2         94800
```

Show

```
##     person_age person_emp_length person_income
## 1           22               123         59000
## 211         21               123        192000
```

4.4 Remove duplicates and invalid values

Show

```
## Cleaning completed.
```

Show

```
## Original rows: 32581
```

Show

```
## Cleaned rows: 32409
```

Show

```
## Columns: 12
```

4.5 Verify the cleaned dataset

Show

```
## [1] 32409    12
```

Show

```
##                 person_age              person_income
##                          0                          0
##      person_home_ownership          person_emp_length
##                          0                        887
##                loan_intent                loan_grade
##                          0                          0
##                  loan_amnt              loan_int_rate
##                          0                       3094
##                loan_status        loan_percent_income
##                          0                          0
##  cb_person_default_on_file cb_person_cred_hist_length
##                          0                          0
```

Show

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   20.00   23.00   26.00   27.73   30.00   94.00
```

Show

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
##   0.000   2.000   4.000   4.783   7.000  41.000     887
```

Show

```
## [1] 0
```

4.6 Convert categorical variables into factors

The target `loan_status` is given the levels `"0"` and `"1"` here, and that order is used consistently in every later chunk.

Show

```
## 'data.frame':    32409 obs. of  12 variables:
##  $ person_age                : int  21 25 23 24 21 26 24 24 21 22 ...
##  $ person_income             : int  9600 9600 65500 54400 9900 77100 78956 83000 10000 850
00 ...
##  $ person_home_ownership     : Factor w/ 4 levels "MORTGAGE","OTHER",..: 3 1 4 4 3 4 4 4 3
4 ...
##  $ person_emp_length         : num  5 1 4 8 2 8 5 8 6 6 ...
##  $ loan_intent               : Factor w/ 6 levels "DEBTCONSOLIDATION",..: 2 4 4 4 6 2 4 5
6 6 ...
##  $ loan_grade                : Factor w/ 7 levels "A","B","C","D",..: 2 3 3 3 1 2 2 1 4 2
...
##  $ loan_amnt                 : int  1000 5500 35000 35000 2500 35000 35000 35000 1600 3500
0 ...
##  $ loan_int_rate             : num  11.14 12.87 15.23 14.27 7.14 ...
##  $ loan_status               : Factor w/ 2 levels "0","1": 1 2 2 2 2 2 2 2 2 2 ...
##  $ loan_percent_income       : num  0.1 0.57 0.53 0.55 0.25 0.45 0.44 0.42 0.16 0.41 ...
##  $ cb_person_default_on_file : Factor w/ 2 levels "N","Y": 1 1 1 2 1 1 1 1 1 1 ...
##  $ cb_person_cred_hist_length: int  2 3 2 4 2 3 4 2 3 4 ...
```

5 Exploratory Data Analysis (EDA)

5.1 Distribution of loan default status

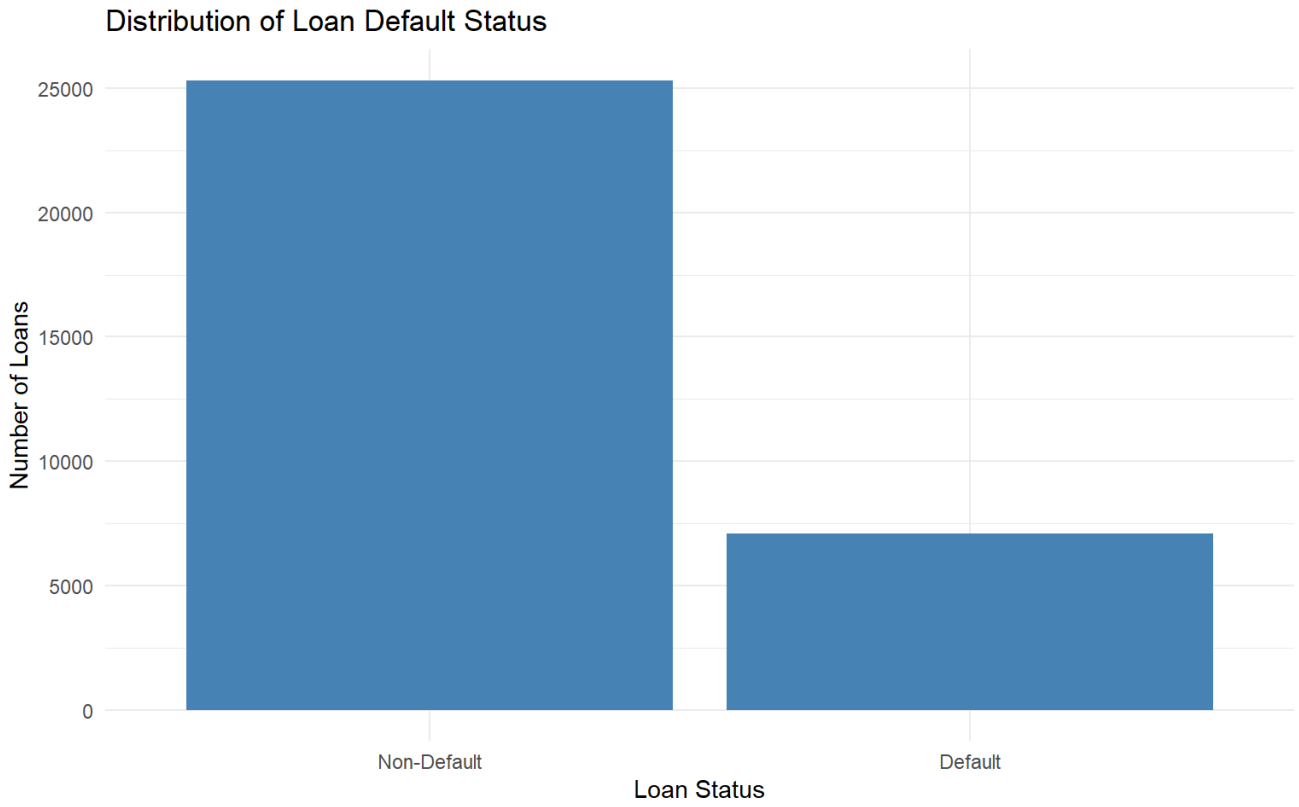
Show

```
##
##     0     1
## 25321  7088
```

Show

```
##
##     0     1
## 78.13 21.87
```

Show

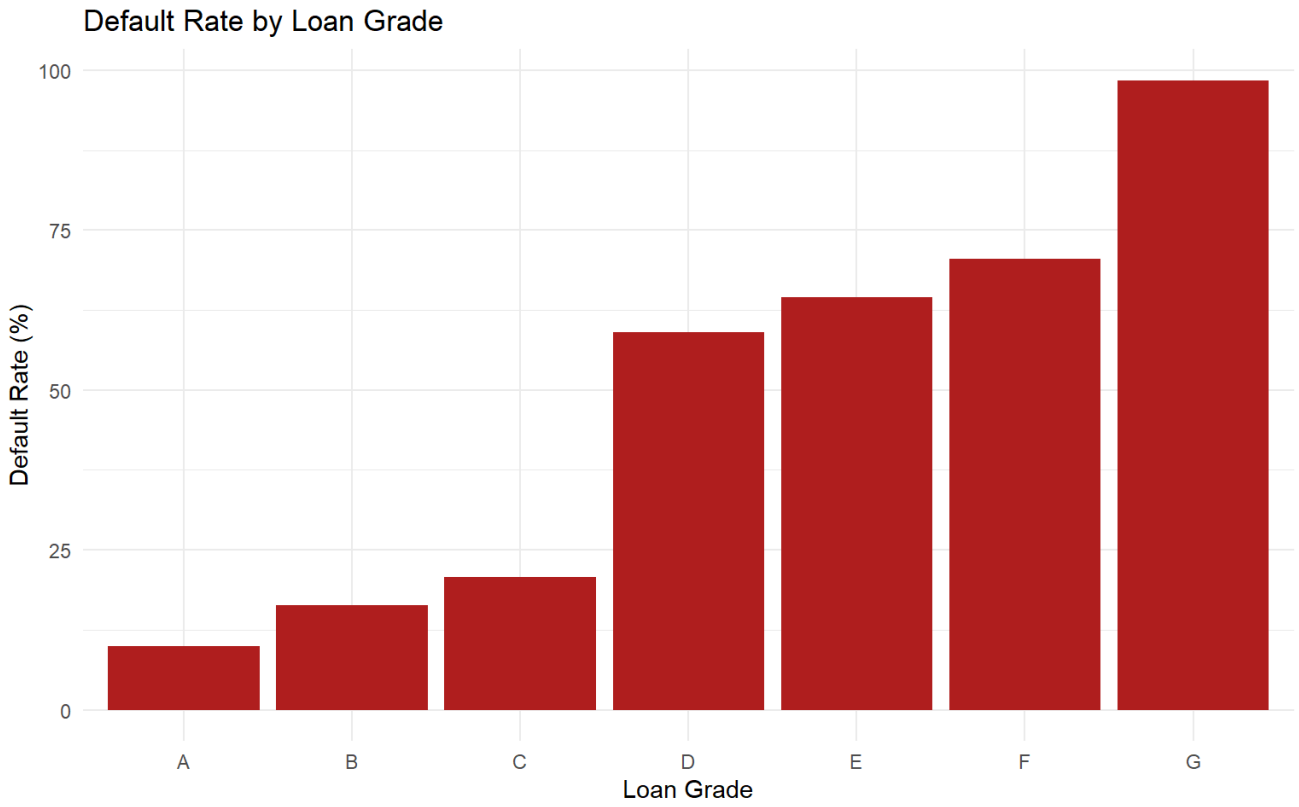


5.2 Default rate by loan grade

Show

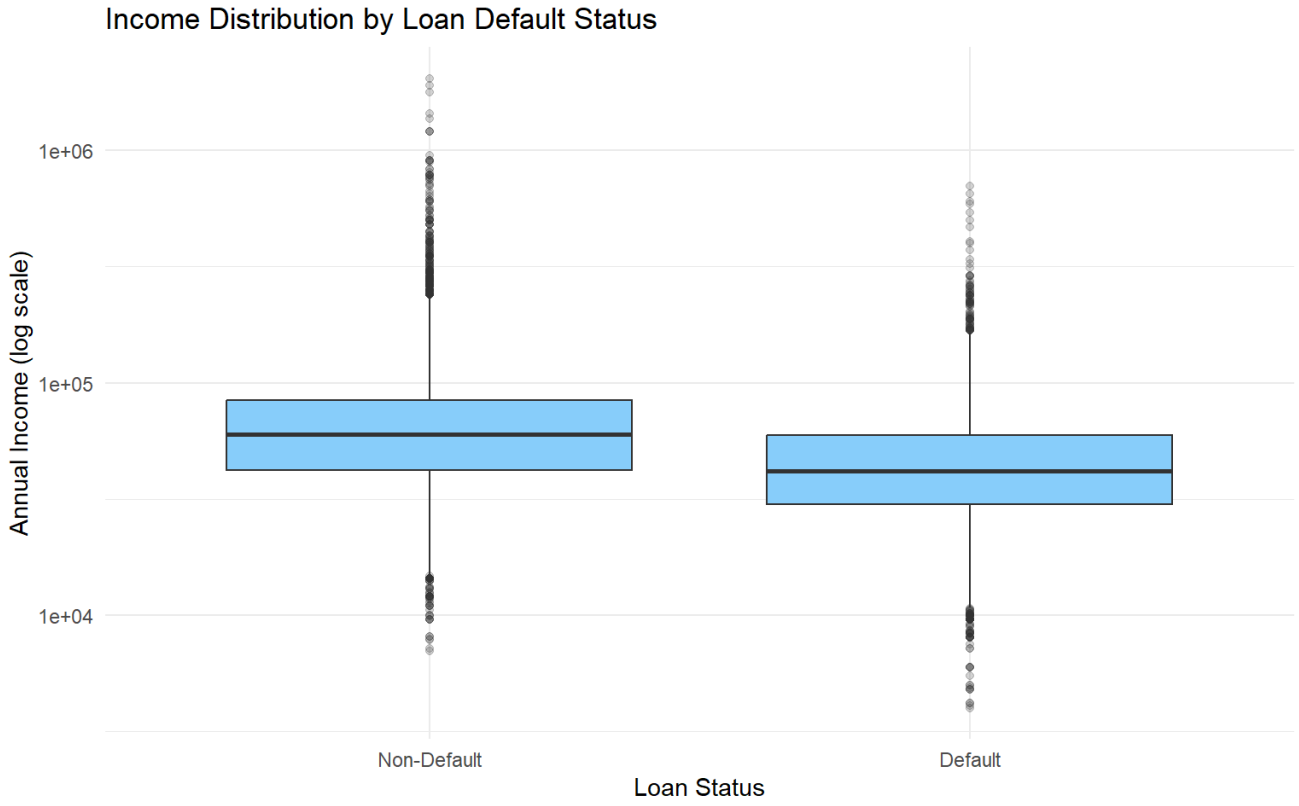
```
## # A tibble: 7 × 4
##   loan_grade Total_Loans Defaults Default_Rate
##   <fct>            <int>    <int>        <dbl>
## 1 A                10702     1066         9.96
## 2 B                10384     1695        16.3
## 3 C                 6436     1336        20.8
## 4 D                 3619     2137        59.0
## 5 E                  963      621        64.5
## 6 F                  241      170        70.5
## 7 G                   64       63        98.4
```

Show



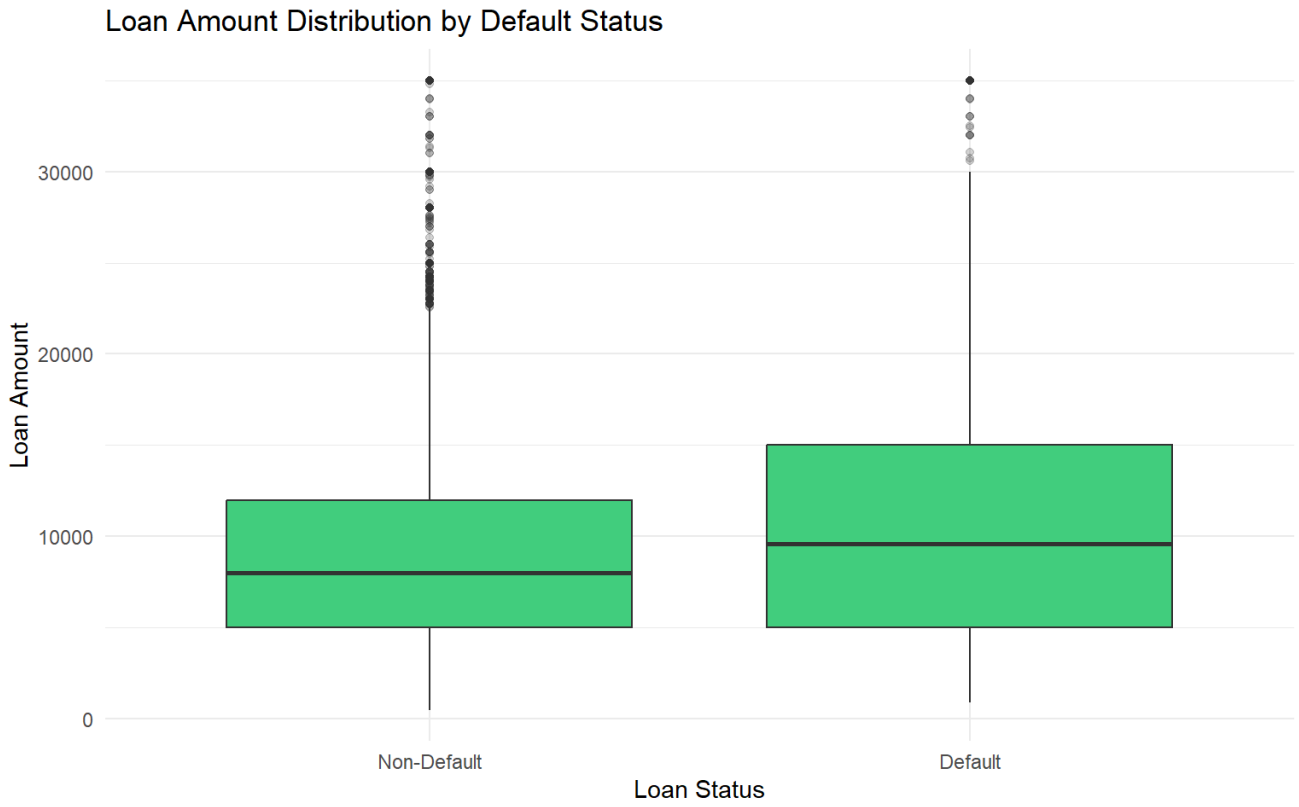
5.3 Income distribution by loan default status

Show



5.4 Loan amount distribution by default status

Show

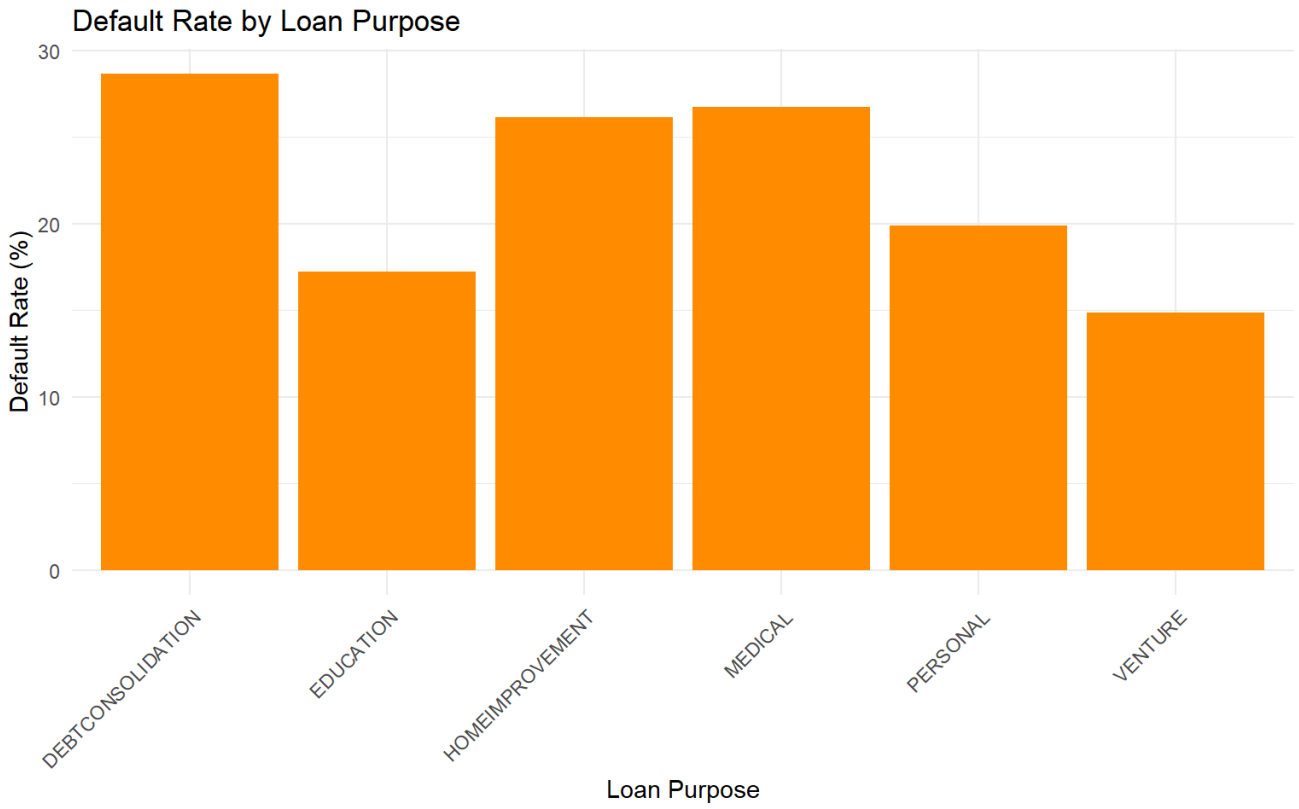


5.5 Default rate by loan purpose

Show

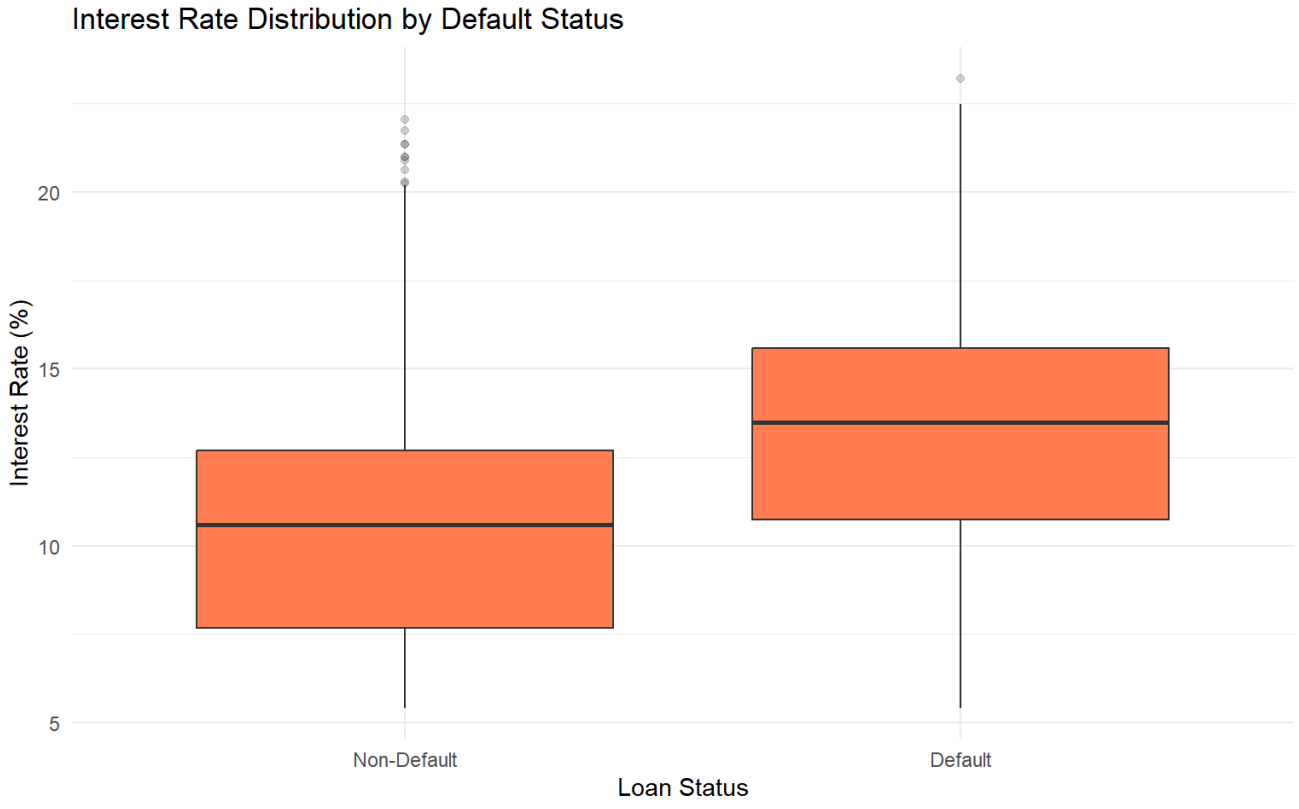
```
## # A tibble: 6 × 4
##   loan_intent       Total_Loans Defaults Default_Rate
##   <fct>                   <int>    <int>        <dbl>
## 1 DEBTCONSOLIDATION        5189     1488         28.7
## 2 EDUCATION                6409     1106         17.3
## 3 HOMEIMPROVEMENT          3594      940         26.2
## 4 MEDICAL                  6042     1617         26.8
## 5 PERSONAL                 5496     1093         19.9
## 6 VENTURE                  5679      844         14.9
```

Show



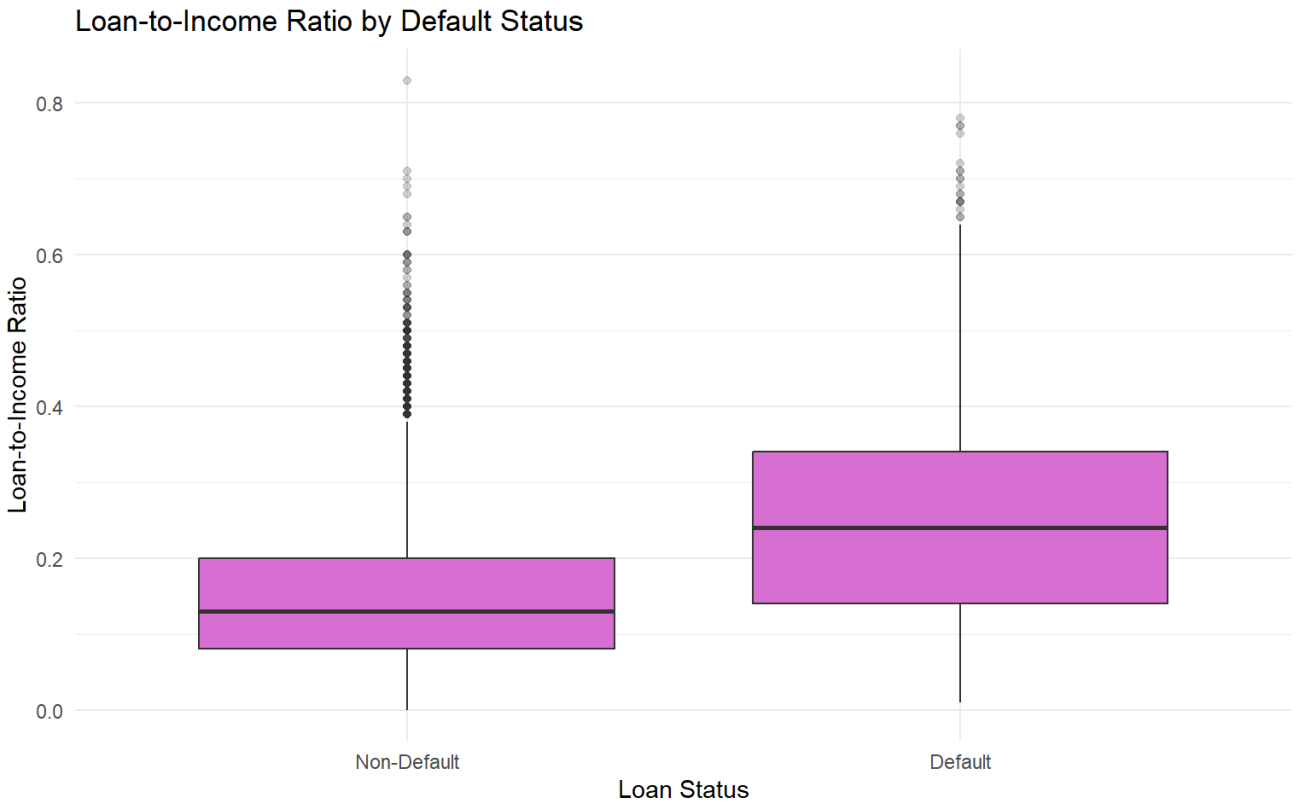
5.6 Interest rate distribution by default status

Show



5.7 Loan-to-income ratio by default status

Show

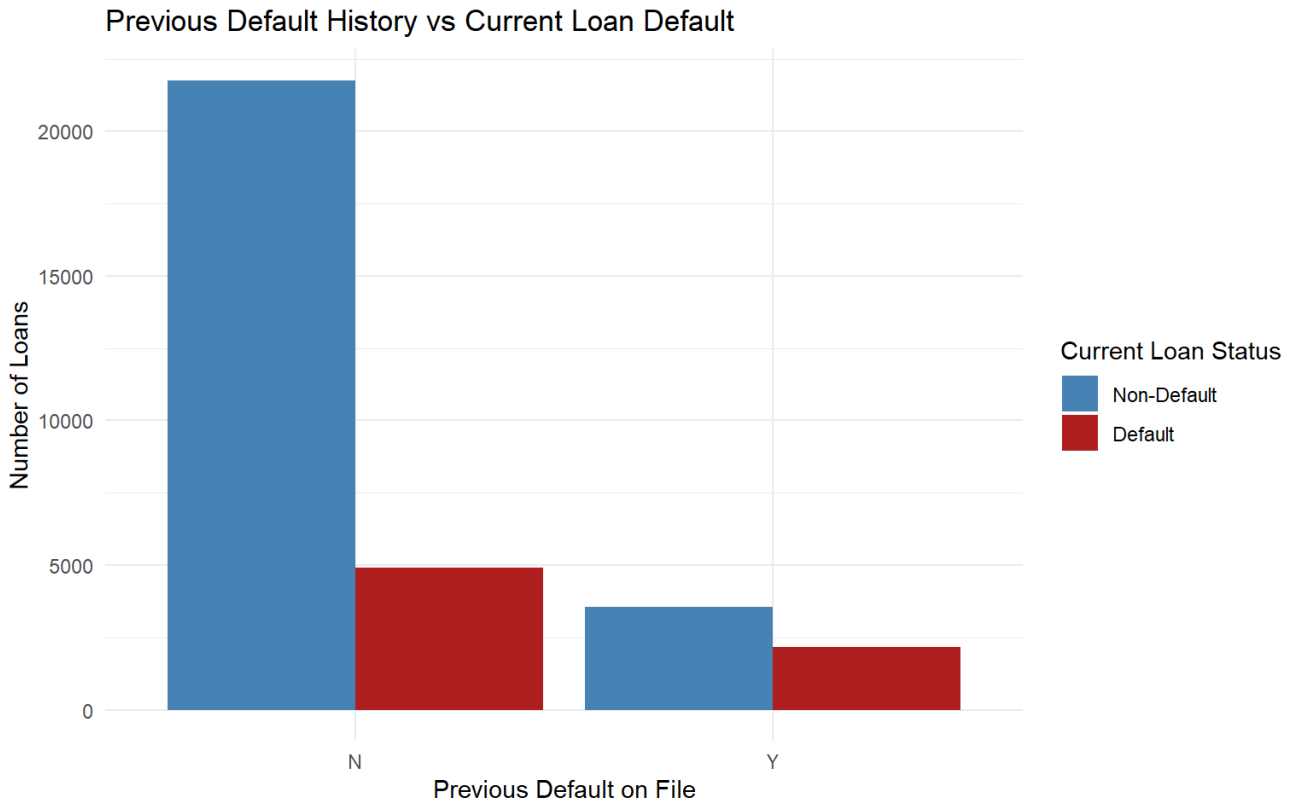


5.8 Previous default history vs current loan default

Show

```
## # A tibble: 4 × 3
##   cb_person_default_on_file loan_status Count
##   <fct>                     <fct>       <int>
## 1 N                         0           21761
## 2 N                         1            4919
## 3 Y                         0            3560
## 4 Y                         1            2169
```

Show

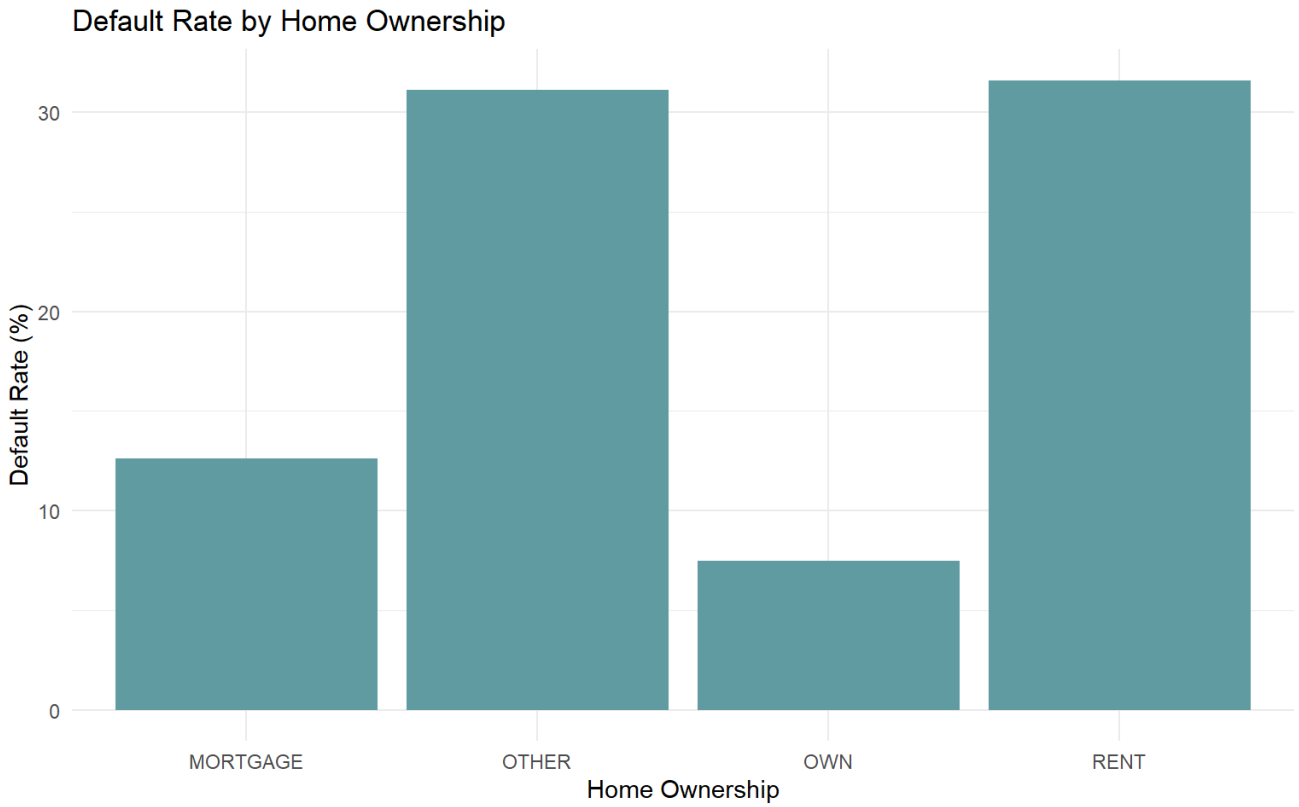


5.9 Default rate by home ownership

Show

```
## # A tibble: 4 × 4
##   person_home_ownership Total_Loans Defaults Default_Rate
##   <fct>                       <int>    <int>        <dbl>
## 1 MORTGAGE                    13366     1687        12.6
## 2 OTHER                         106       33        31.1
## 3 OWN                          2563      192         7.49
## 4 RENT                        16374     5176        31.6
```

Show

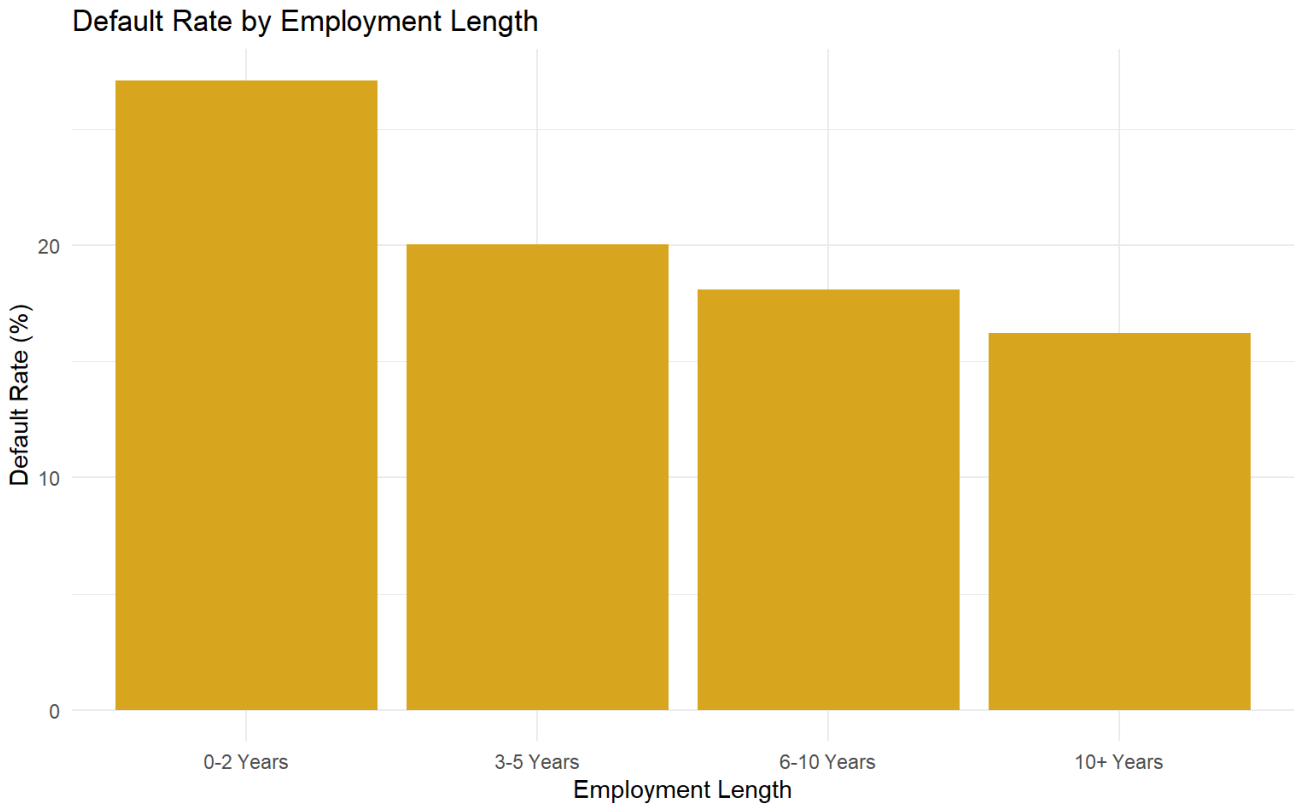


5.10 Default rate by employment length

Show

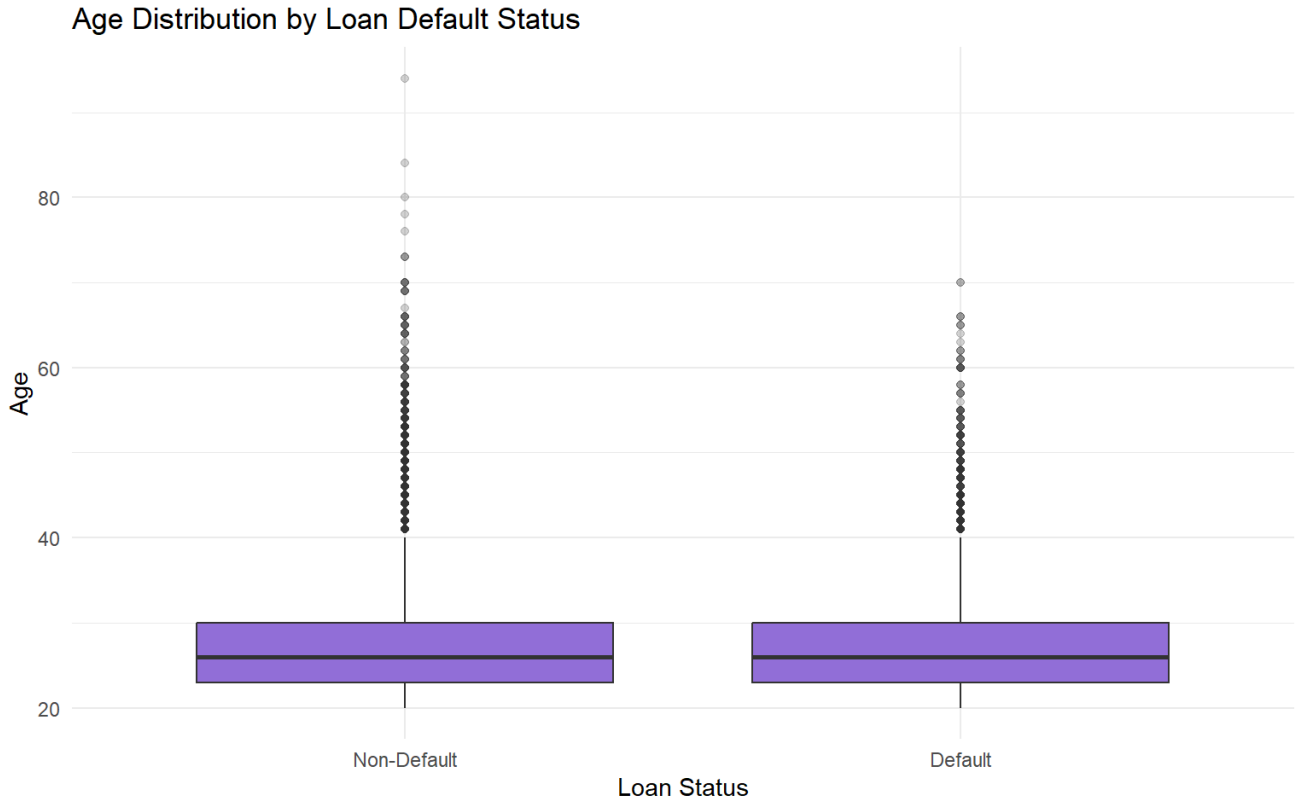
```
## # A tibble: 4 × 4
##   Employment_Group Total_Loans Defaults Default_Rate
##   <fct>                 <int>    <int>        <dbl>
## 1 0-2 Years             10814     2933         27.1
## 2 3-5 Years              9229     1851         20.1
## 3 6-10 Years             8564     1550         18.1
## 4 10+ Years              2915      473         16.2
```

Show



5.11 Age distribution by loan default status

Show

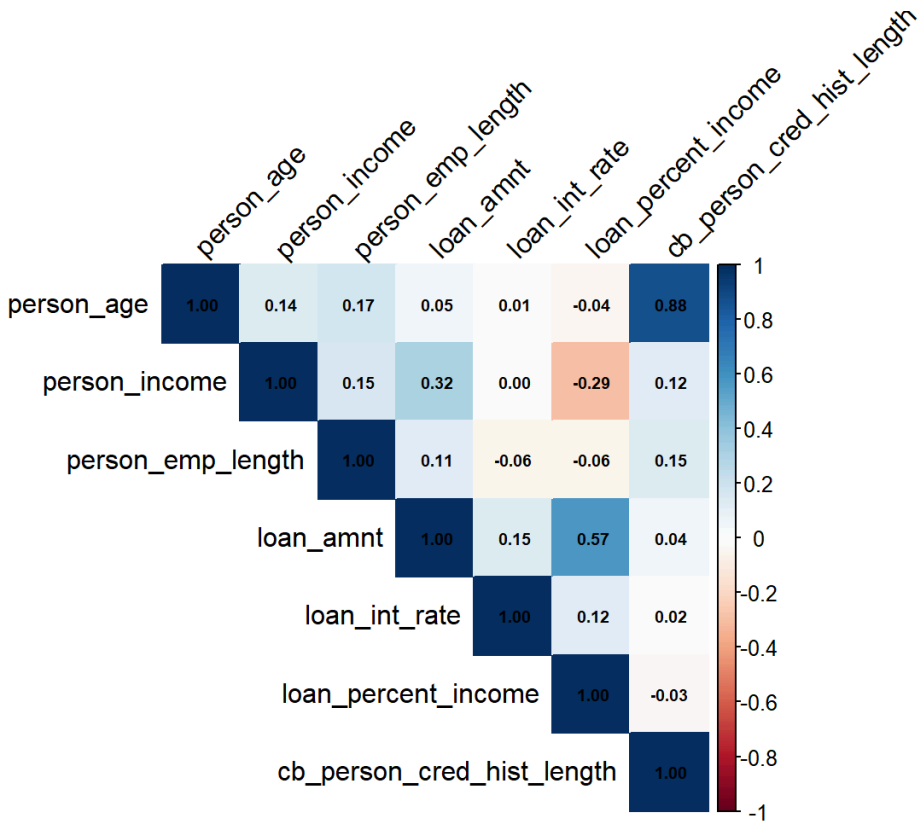


5.12 Correlation matrix of numerical variables

Show

```
##                             person_age person_income person_emp_length loan_amnt
## person_age                       1.000         0.140             0.172     0.052
## person_income                    0.140         1.000             0.154     0.317
## person_emp_length                0.172         0.154             1.000     0.111
## loan_amnt                        0.052         0.317             0.111     1.000
## loan_int_rate                    0.011        -0.001            -0.058     0.147
## loan_percent_income             -0.041        -0.295            -0.058     0.573
## cb_person_cred_hist_length       0.878         0.123             0.150     0.042
##                             loan_int_rate loan_percent_income
## person_age                          0.011              -0.041
## person_income                      -0.001              -0.295
## person_emp_length                  -0.058              -0.058
## loan_amnt                           0.147               0.573
## loan_int_rate                       1.000               0.120
## loan_percent_income                 0.120               1.000
## cb_person_cred_hist_length          0.016              -0.031
##                             cb_person_cred_hist_length
## person_age                                       0.878
## person_income                                    0.123
## person_emp_length                                0.150
## loan_amnt                                        0.042
## loan_int_rate                                    0.016
## loan_percent_income                             -0.031
## cb_person_cred_hist_length                       1.000
```

Show



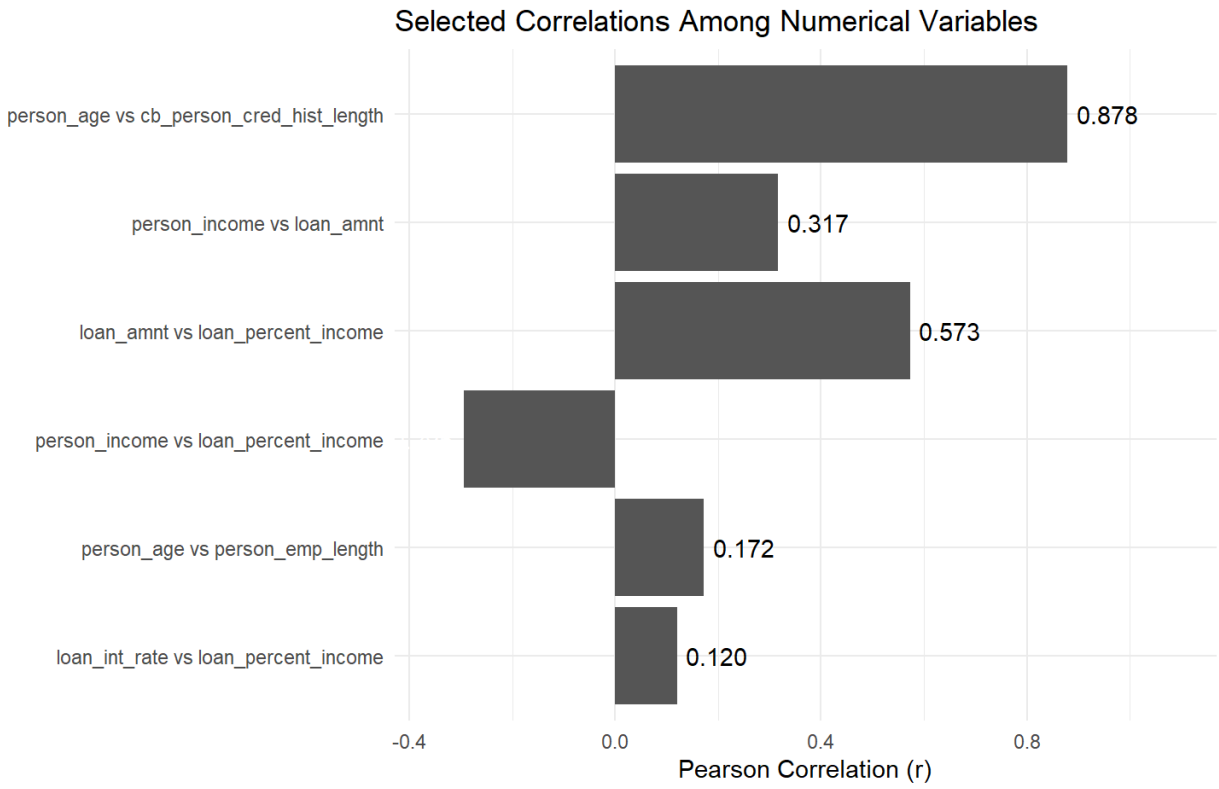
5.13 Selected correlations among numerical variables

The notebook plotted an object called `correlations` without a visible definition. It is defined here from `cor_matrix` (computed above) for a set of variable pairs, so the plot always reflects the data. The pairs shown are the ones selected for this report.

Show

```
##                              Relationship Correlation
## 1 person_age vs cb_person_cred_hist_length   0.8782405
## 2               person_income vs loan_amnt   0.3167319
## 3         loan_amnt vs loan_percent_income   0.5726316
## 4     person_income vs loan_percent_income  -0.2947660
## 5          person_age vs person_emp_length   0.1724812
## 6     loan_int_rate vs loan_percent_income   0.1201111
```

Show



6 Statistical Analysis and Data Preparation

6.1 Missing-value treatment: preliminary median imputation

Two variables contain missing values: `person_emp_length` and `loan_int_rate`. This chunk fills them with the **full-data** median on a separate copy (`loan_imputed`) so that descriptive statistics and the chi-square tests can use every row. It is **not** used for modelling: the models use training-set medians only (see the Logistic Regression section) to avoid information leaking from the test set.

Show

```
## Missing values before treatment:
```

Show

```
##                 person_age              person_income
##                          0                          0
##      person_home_ownership          person_emp_length
##                          0                        887
##                loan_intent                 loan_grade
##                          0                          0
##                  loan_amnt              loan_int_rate
##                          0                       3094
##                loan_status        loan_percent_income
##                          0                          0
##  cb_person_default_on_file cb_person_cred_hist_length
##                          0                          0
```

Show

```
##
## Missing values after treatment:
```

Show

```
##                 person_age              person_income
##                          0                          0
##      person_home_ownership          person_emp_length
##                          0                          0
##                loan_intent                 loan_grade
##                          0                          0
##                  loan_amnt              loan_int_rate
##                          0                          0
##                loan_status        loan_percent_income
##                          0                          0
##  cb_person_default_on_file cb_person_cred_hist_length
##                          0                          0
```

Show

```
##
## Dataset dimensions after imputation:
```

Show

```
## [1] 32409    12
```

Show

```
##
## Median employment length used: 4
```

Show

```
## Median interest rate used: 10.99
```

Show

```
## Duplicate rows: 0
```

Show

```
## 'data.frame':    32409 obs. of  12 variables:
##  $ person_age                : int  21 25 23 24 21 26 24 24 21 22 ...
##  $ person_income             : int  9600 9600 65500 54400 9900 77100 78956 83000 10000 850
00 ...
##  $ person_home_ownership     : Factor w/ 4 levels "MORTGAGE","OTHER",..: 3 1 4 4 3 4 4 4 3
4 ...
##  $ person_emp_length         : num  5 1 4 8 2 8 5 8 6 6 ...
##  $ loan_intent               : Factor w/ 6 levels "DEBTCONSOLIDATION",..: 2 4 4 4 6 2 4 5
6 6 ...
##  $ loan_grade                : Factor w/ 7 levels "A","B","C","D",..: 2 3 3 3 1 2 2 1 4 2
...
##  $ loan_amnt                 : int  1000 5500 35000 35000 2500 35000 35000 35000 1600 3500
0 ...
##  $ loan_int_rate             : num  11.14 12.87 15.23 14.27 7.14 ...
##  $ loan_status               : Factor w/ 2 levels "0","1": 1 2 2 2 2 2 2 2 2 2 ...
##  $ loan_percent_income       : num  0.1 0.57 0.53 0.55 0.25 0.45 0.44 0.42 0.16 0.41 ...
##  $ cb_person_default_on_file : Factor w/ 2 levels "N","Y": 1 1 1 2 1 1 1 1 1 1 ...
##  $ cb_person_cred_hist_length: int  2 3 2 4 2 3 4 2 3 4 ...
```

Show

```
##    person_age    person_income     person_home_ownership person_emp_length
##  Min.   :20.00   Min.   :   4000   MORTGAGE:13366        Min.   : 0.000
##  1st Qu.:23.00   1st Qu.:  38500   OTHER   :  106        1st Qu.: 2.000
##  Median :26.00   Median :  55000   OWN     : 2563        Median : 4.000
##  Mean   :27.73   Mean   :  65894   RENT    :16374        Mean   : 4.761
##  3rd Qu.:30.00   3rd Qu.:  79200                         3rd Qu.: 7.000
##  Max.   :94.00   Max.   :2039784                         Max.   :41.000
##
##             loan_intent   loan_grade   loan_amnt     loan_int_rate
##  DEBTCONSOLIDATION:5189   A:10702    Min.   :  500   Min.   : 5.42
##  EDUCATION        :6409   B:10384    1st Qu.: 5000   1st Qu.: 8.49
##  HOMEIMPROVEMENT  :3594   C: 6436    Median : 8000   Median :10.99
##  MEDICAL          :6042   D: 3619    Mean   : 9592   Mean   :11.01
##  PERSONAL         :5496   E:  963    3rd Qu.:12250   3rd Qu.:13.11
##  VENTURE          :5679   F:  241    Max.   :35000   Max.   :23.22
##                           G:   64
##  loan_status loan_percent_income cb_person_default_on_file
##  0:25321     Min.   :0.0000      N:26680
##  1: 7088     1st Qu.:0.0900      Y: 5729
##              Median :0.1500
##              Mean   :0.1702
##              3rd Qu.:0.2300
##              Max.   :0.8300
##
##  cb_person_cred_hist_length
##  Min.   : 2.000
##  1st Qu.: 3.000
##  Median : 4.000
##  Mean   : 5.811
##  3rd Qu.: 8.000
##  Max.   :30.000
##
```

Show

```
##
## Missing-value treatment completed successfully.
```

6.2 Chi-square tests of association

Each categorical predictor is tested against `loan_status`. A small p-value indicates an association, not causation. Some cells (for example loan grade G and home ownership OTHER) are small, so those results should be read with care.

Show

```
##                   Variable Chi_Square Degrees_of_Freedom    P_Value
##                 loan_grade   5584.784                  6  0.000e+00
##                loan_intent    519.625                  5 4.632e-110
##      person_home_ownership   1893.770                  3  0.000e+00
##  cb_person_default_on_file   1040.119                  1 3.419e-228
##                             Conclusion
##  Statistically significant association
##  Statistically significant association
##  Statistically significant association
##  Statistically significant association
```

6.3 Train-test split: 80:20 stratified sampling

The split is made on `loan_clean` (before any imputation), so no test-set information is used to fill missing values.

Show

```
## Training rows: 25928
```

Show

```
## Testing rows: 6481
```

Show

```
##
## Training target distribution (%):
```

Show

```
##
##     0     1
## 78.13 21.87
```

Show

```
##
## Testing target distribution (%):
```

Show

```
##
##     0     1
## 78.14 21.86
```

Show

```
##
## Train/test split completed.
```

7 Machine Learning Models

7.1 Prepare data and train Logistic Regression

Missing values are filled with medians calculated from the **training** data only, and the same values are applied to the test data.

Show

```
##
## Call:
## glm(formula = loan_status ~ ., family = binomial(link = "logit"),
##     data = train_data)
##
## Coefficients:
##                               Estimate Std. Error z value Pr(>|z|)
## (Intercept)                 -4.099e+00  1.945e-01 -21.072  < 2e-16 ***
## person_age                  -4.163e-03  6.561e-03  -0.635  0.52575
## person_income                1.640e-06  5.962e-07   2.751  0.00595 **
## person_home_ownershipOTHER   7.714e-01  3.194e-01   2.415  0.01573 *
## person_home_ownershipOWN    -1.787e+00  1.141e-01 -15.669  < 2e-16 ***
## person_home_ownershipRENT    8.091e-01  4.479e-02  18.064  < 2e-16 ***
## person_emp_length           -1.432e-02  5.376e-03  -2.663  0.00774 **
## loan_intentEDUCATION        -8.743e-01  6.412e-02 -13.636  < 2e-16 ***
## loan_intentHOMEIMPROVEMENT   4.334e-02  7.093e-02   0.611  0.54119
## loan_intentMEDICAL          -1.857e-01  6.042e-02  -3.074  0.00212 **
## loan_intentPERSONAL         -6.458e-01  6.556e-02  -9.850  < 2e-16 ***
## loan_intentVENTURE          -1.107e+00  6.896e-02 -16.060  < 2e-16 ***
## loan_gradeB                  1.777e-01  7.114e-02   2.498  0.01247 *
## loan_gradeC                  4.859e-01  1.010e-01   4.812 1.49e-06 ***
## loan_gradeD                  2.532e+00  1.237e-01  20.470  < 2e-16 ***
## loan_gradeE                  2.759e+00  1.608e-01  17.155  < 2e-16 ***
## loan_gradeF                  3.278e+00  2.387e-01  13.734  < 2e-16 ***
## loan_gradeG                  6.523e+00  1.049e+00   6.215 5.12e-10 ***
## loan_amnt                   -1.076e-04  5.118e-06 -21.021  < 2e-16 ***
## loan_int_rate                6.182e-02  1.438e-02   4.300 1.71e-05 ***
## loan_percent_income          1.352e+01  2.961e-01  45.657  < 2e-16 ***
## cb_person_default_on_fileY -2.087e-02  5.579e-02  -0.374  0.70839
## cb_person_cred_hist_length -2.369e-03  1.001e-02  -0.237  0.81288
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 27239  on 25927  degrees of freedom
## Residual deviance: 17530  on 25905  degrees of freedom
## AIC: 17576
##
## Number of Fisher Scoring iterations: 6
```

Show

```
##
## Logistic Regression training completed.
```

Show

```
## Training rows: 25928
```

Show

```
## Testing rows: 6481
```

Show

```
## Remaining missing values in training data: 0
```

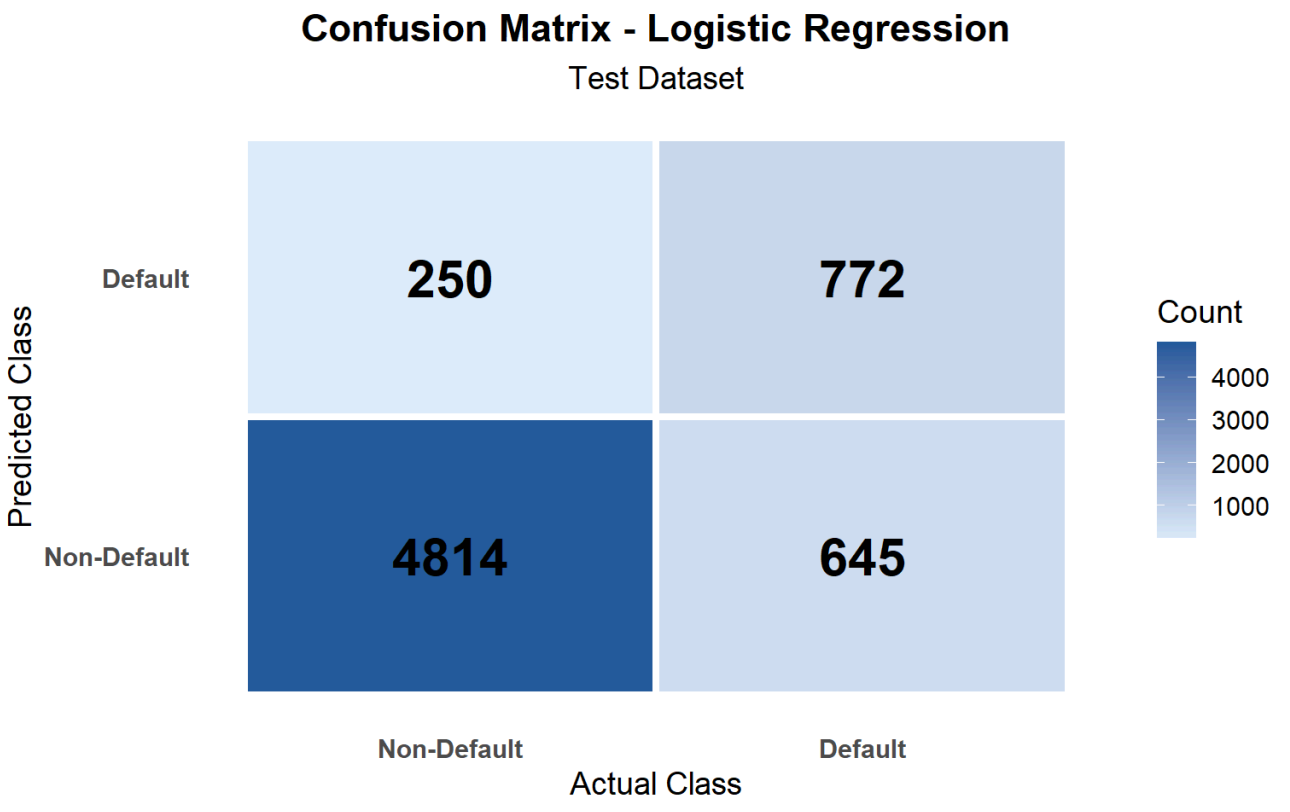
Show

```
## Remaining missing values in testing data: 0
```

7.2 Evaluate Logistic Regression on test data

`actual_status` and `conf_matrix` are created first, and the confusion-matrix plot and metrics are built from them. In the confusion table, rows are predictions and columns are actual classes, with `"1"` (default) as the positive class.

Show



Show

```
##
## --- Logistic Regression Test Metrics ---
```

Show

```
## Accuracy : 0.8619
```

Show

```
## Precision: 0.7554
```

Show

```
## Recall   : 0.5448
```

Show

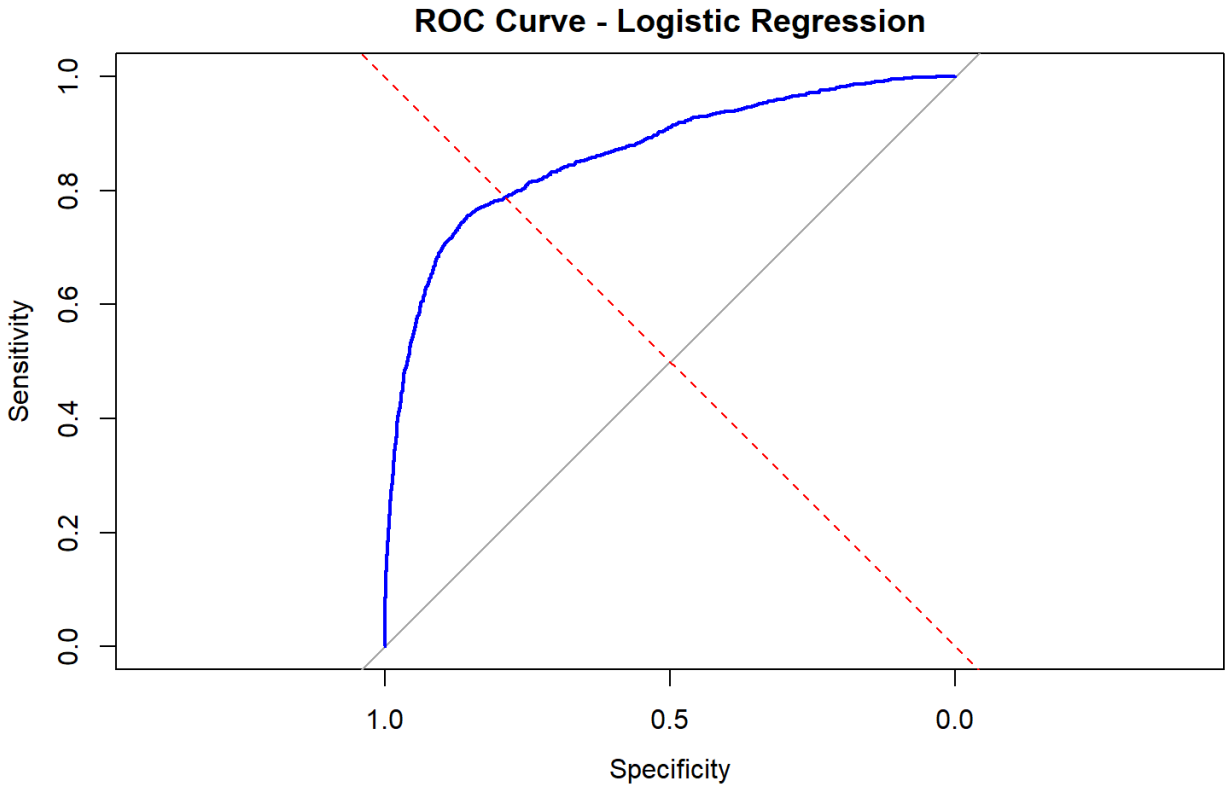
```
## F1-score : 0.633
```

7.3 ROC curve and AUC: Logistic Regression

Show

```
## Logistic Regression ROC-AUC: 0.8639
```

Show



7.4 Train the Random Forest model

Show

```
##
## Call:
##  randomForest(formula = loan_status ~ ., data = train_data, ntree = 300,      importance =
TRUE)
##                Type of random forest: classification
##                      Number of trees: 300
## No. of variables tried at each split: 3
##
##         OOB estimate of  error rate: 6.63%
## Confusion matrix:
##       0    1 class.error
## 0 20147  110 0.005430222
## 1  1608 4063 0.283547875
```

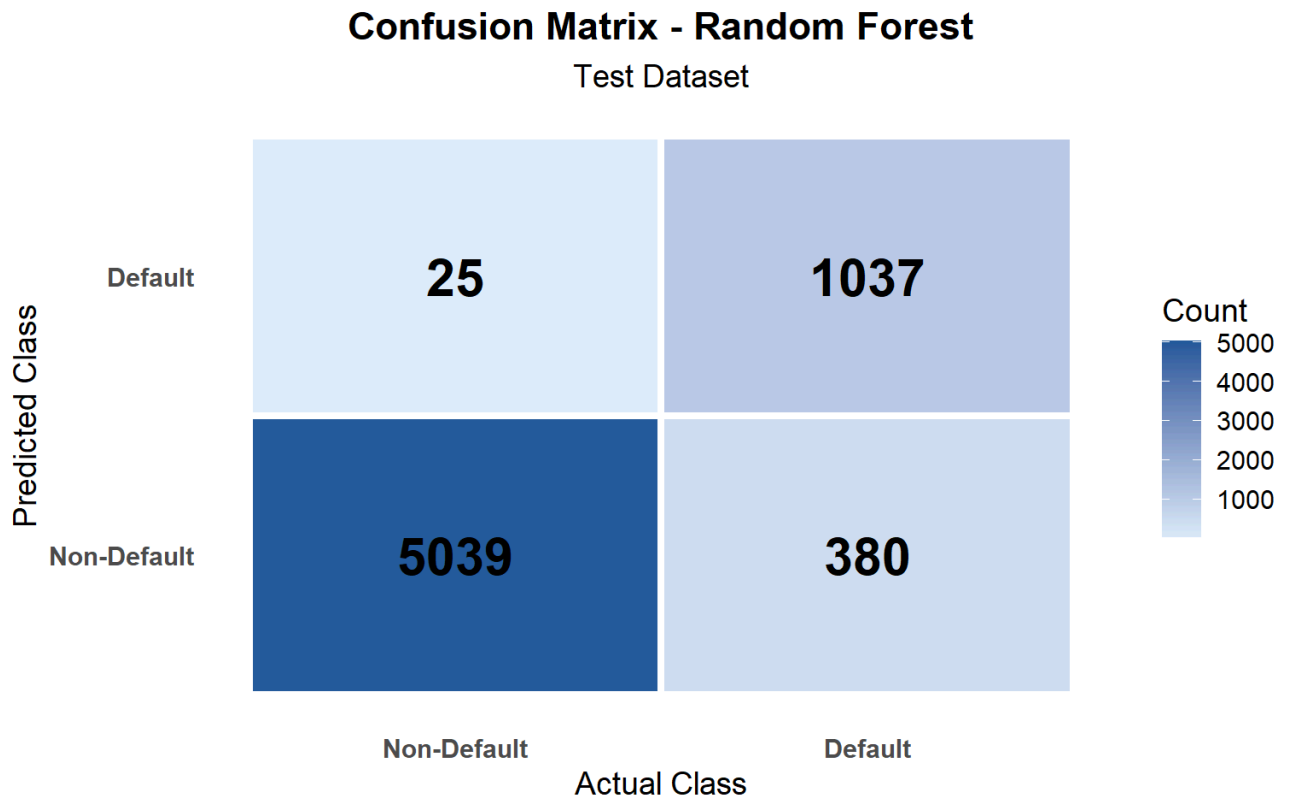
Show

```
##
## Random Forest training completed.
```

7.5 Evaluate Random Forest on test data

As with Logistic Regression, `rf_conf_matrix` is created before it is plotted or used for metrics. The predicted classes (`rf_pred`) come from the forest's majority vote, which can differ very slightly from applying a 0.50 cut-off to `rf_prob` (this is why the threshold table later shows a tiny difference at 0.50).

Show



Show

```
##
## --- Random Forest Test Metrics ---
```

Show

```
## Accuracy : 0.9375
```

Show

```
## Precision: 0.9765
```

Show

```
## Recall   : 0.7318
```

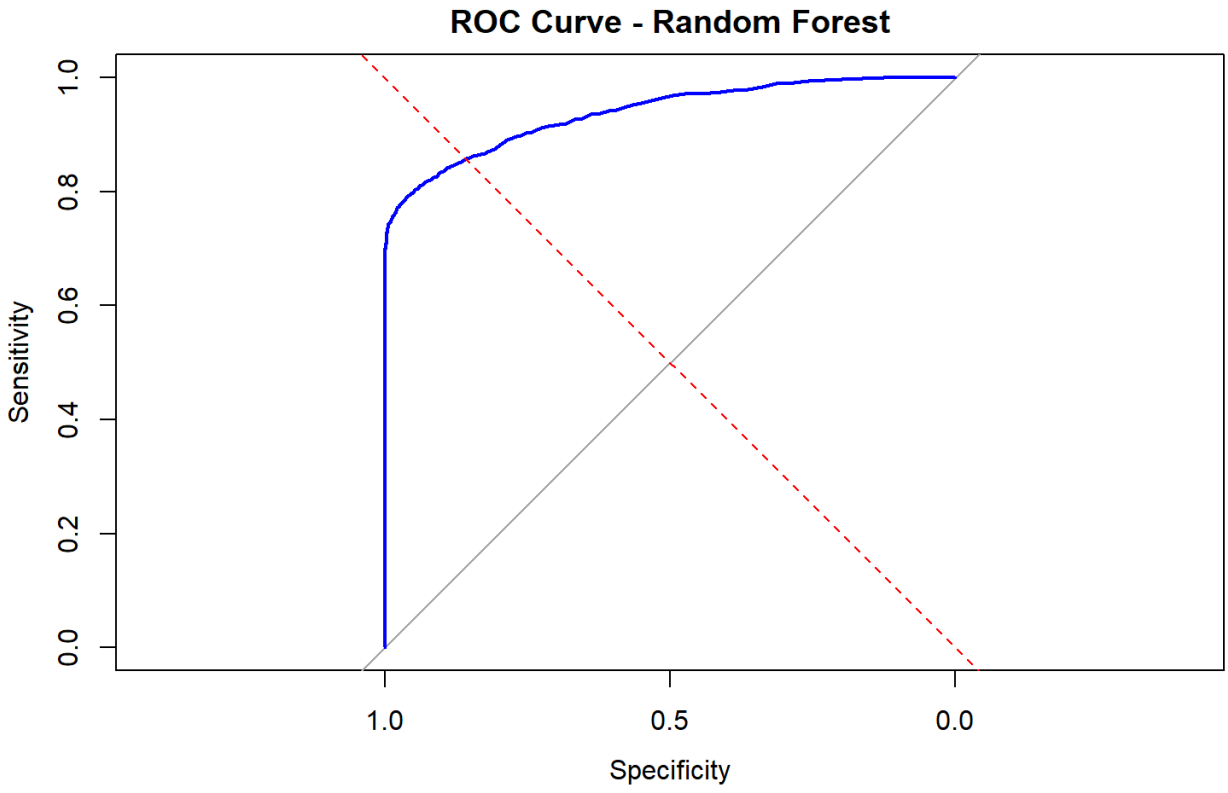
Show

```
## F1-score : 0.8366
```

Show

```
## ROC-AUC  : 0.9384
```

Show

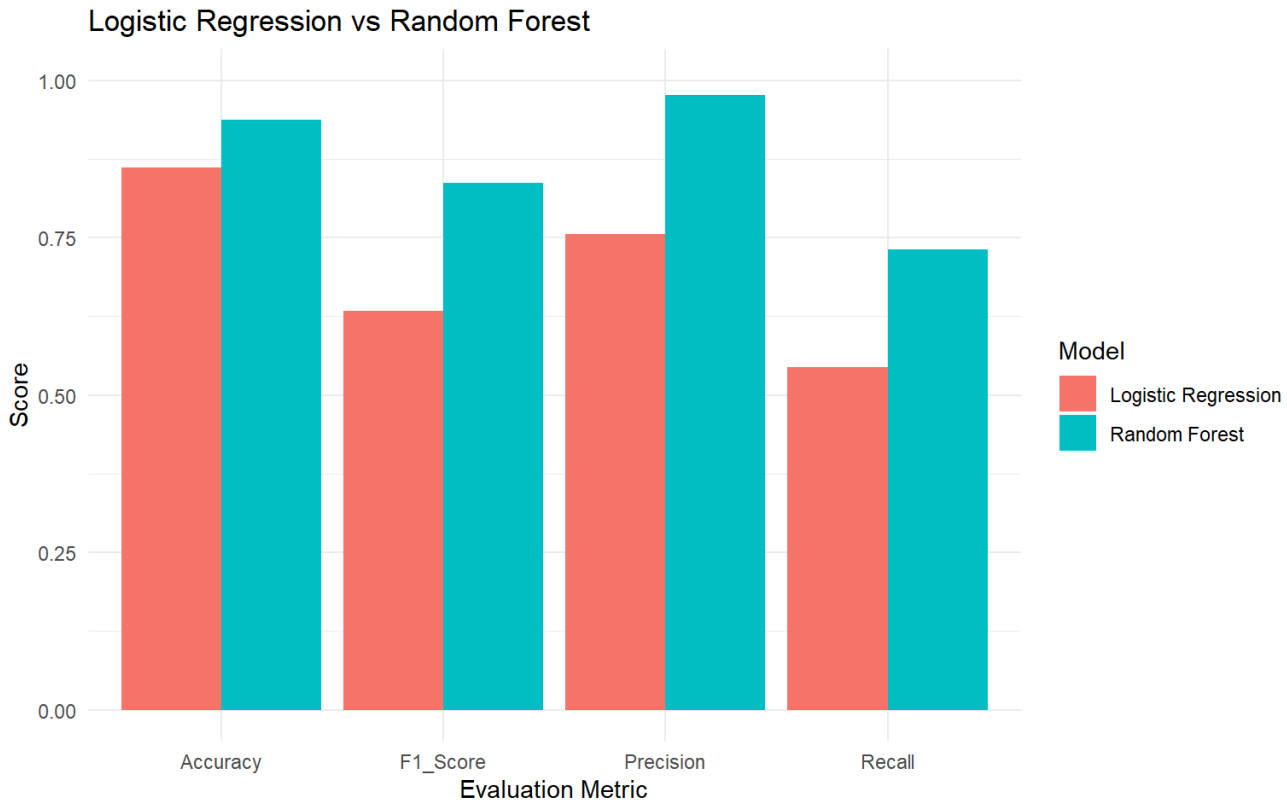


7.6 Model performance comparison

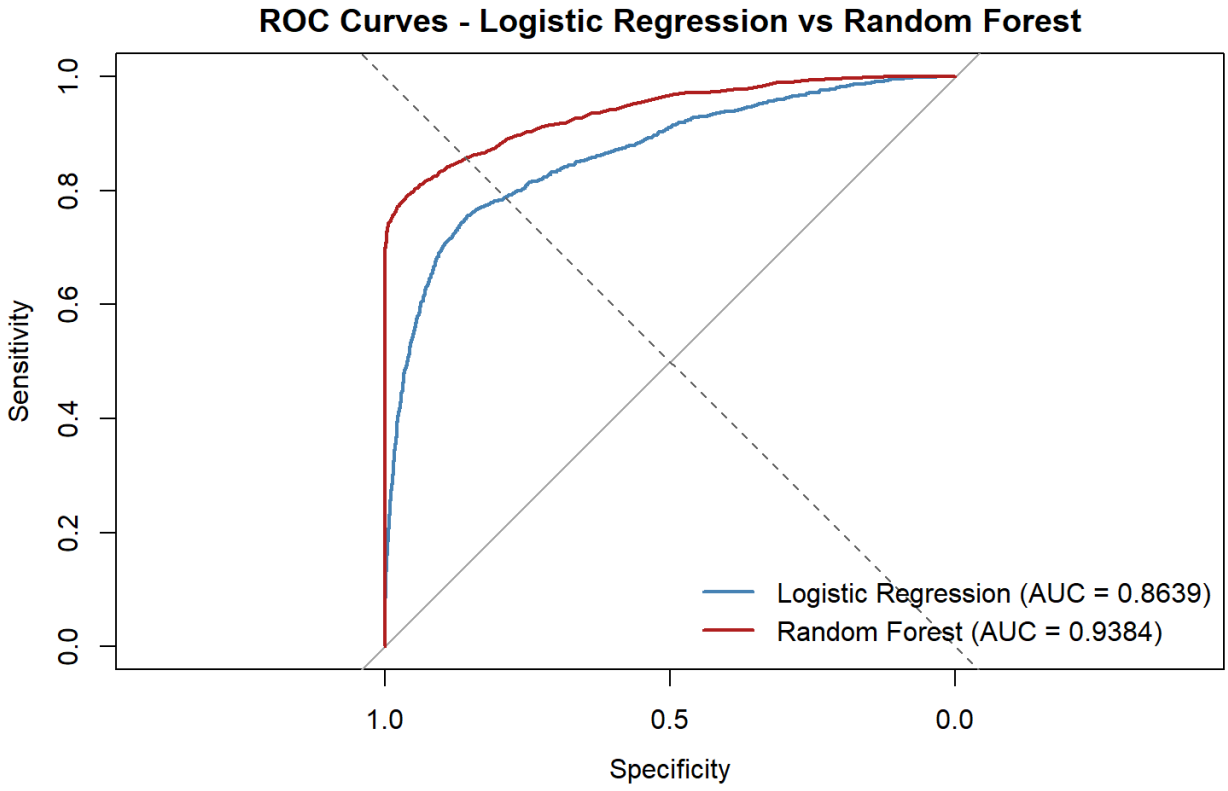
Show

```
##               Model Accuracy Precision Recall F1_Score ROC_AUC
##  Logistic Regression   0.8619    0.7554 0.5448   0.6330  0.8639
##        Random Forest   0.9375    0.9765 0.7318   0.8366  0.9384
```

Show



Show

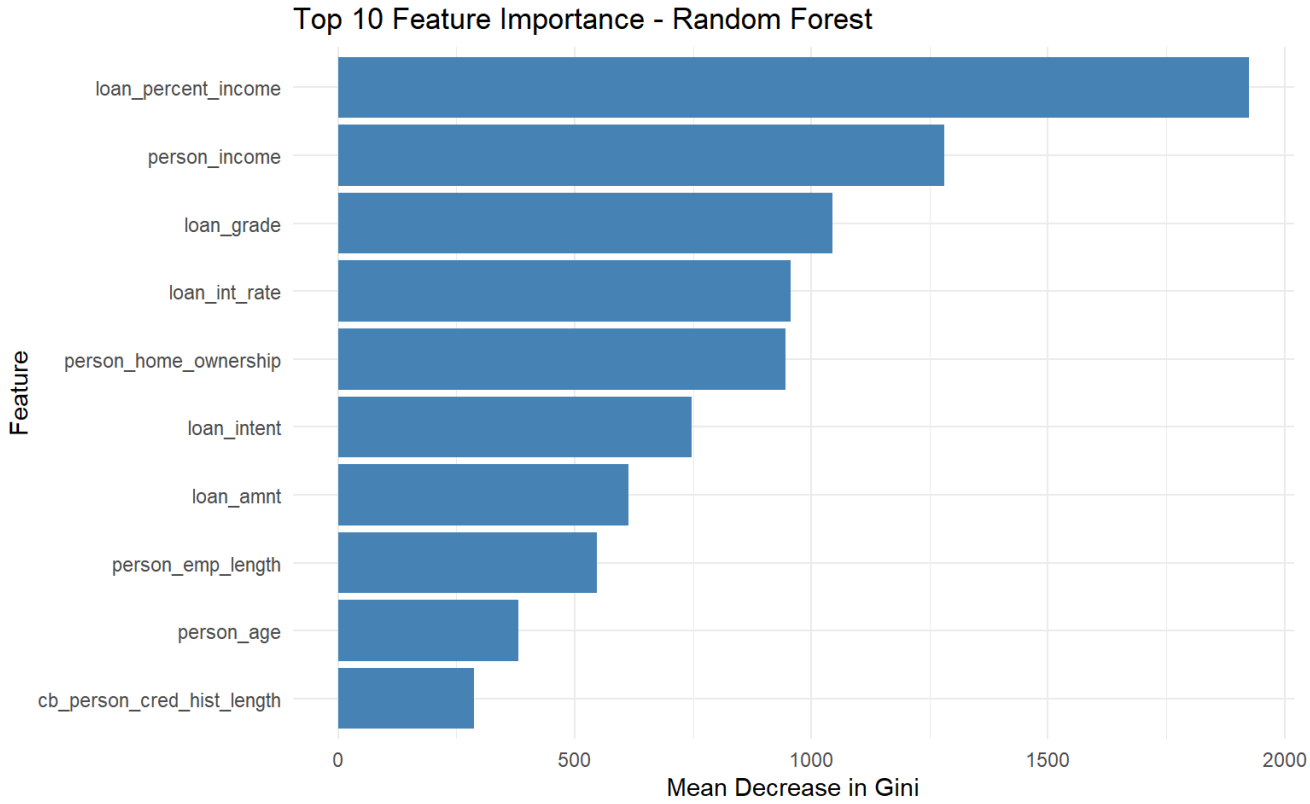


7.7 Feature importance of the Random Forest

Show

```
##                     Feature Importance
##         loan_percent_income  1924.0951
##               person_income  1280.4665
##                  loan_grade  1044.5926
##               loan_int_rate   957.0813
##       person_home_ownership   944.5979
##                 loan_intent   747.2400
##                   loan_amnt   613.5121
##           person_emp_length   547.3614
##                  person_age   381.5152
##  cb_person_cred_hist_length   287.6509
##   cb_person_default_on_file    80.5233
```

Show



Feature importance shows which variables the forest relied on most; it does not show that a variable causes default.

7.8 Final model validation checks

Show

```
## ===== FINAL PROJECT VALIDATION =====
```

Show

```
## Training rows: 25928
```

Show

```
## Testing rows: 6481
```

Show

```
## Missing values in training data: 0
```

Show

```
## Missing values in testing data: 0
```

Show

```
##
## Test prediction count: 6481
```

Show

```
## Actual test labels: 6481
```

Show

```
##
## ===== RANDOM FOREST RESULTS (threshold from majority vote) =====
```

Show

```
## Accuracy : 93.75 %
```

Show

```
## Precision: 97.65 %
```

Show

```
## Recall   : 73.18 %
```

Show

```
## F1-score : 0.8366
```

Show

```
## ROC-AUC  : 0.9384
```

Show

```
##
## All final validation checks passed.
```

8 Experiments

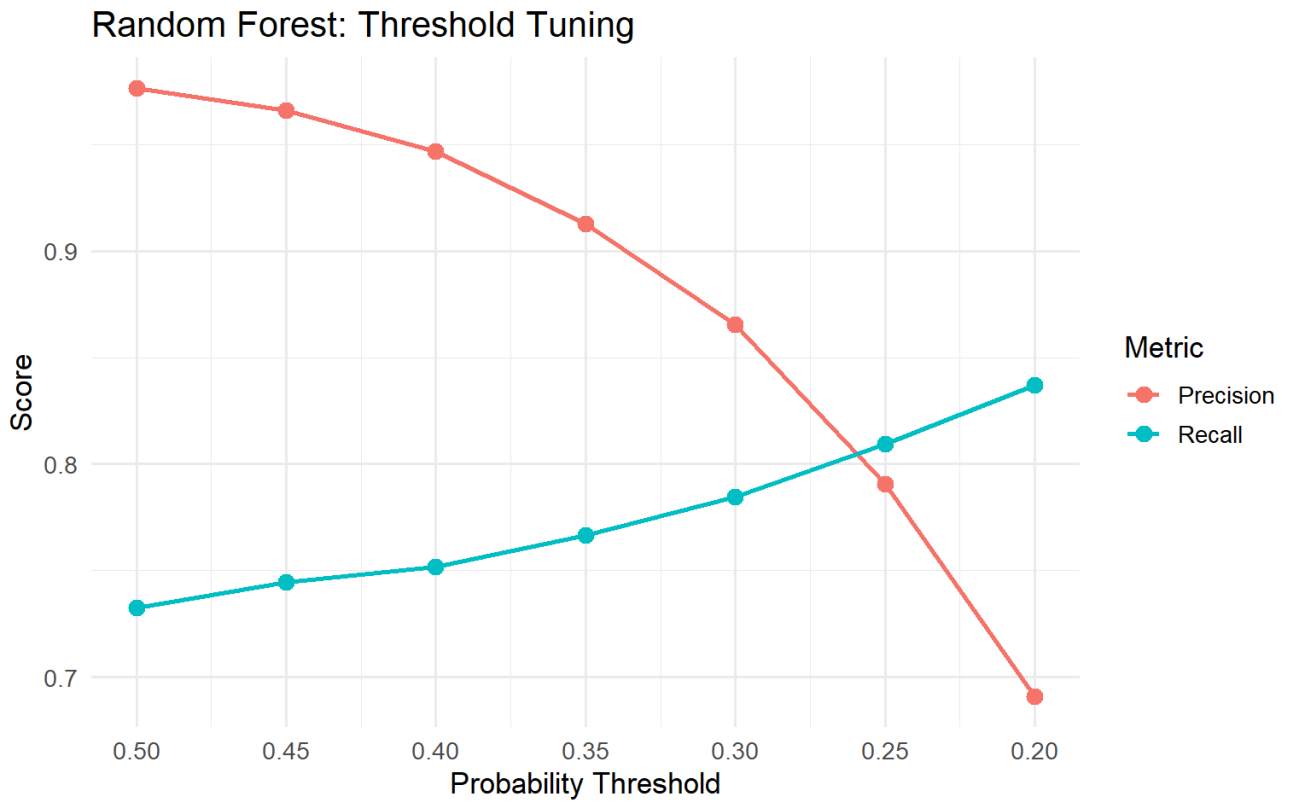
8.1 Random Forest threshold-tuning experiment

The default decision threshold is 0.50. Lowering it flags more loans as defaults, which usually raises recall and lowers precision.

Show

```
##  Threshold Accuracy Precision Recall     F1
##       0.50   0.9377    0.9765 0.7325 0.8371
##       0.45   0.9384    0.9661 0.7445 0.8410
##       0.40   0.9364    0.9467 0.7516 0.8379
##       0.35   0.9329    0.9126 0.7664 0.8331
##       0.30   0.9262    0.8654 0.7848 0.8231
##       0.25   0.9114    0.7905 0.8095 0.7999
##       0.20   0.8824    0.6907 0.8370 0.7569
```

Show



Limitation: the threshold of 0.40 was chosen after looking at the **test-set** results above. The 0.40 figures are therefore an experiment, not an unbiased estimate of future performance. A stricter design would choose the threshold on a separate validation set and evaluate it once on an untouched test set.

8.2 Class-weighted Random Forest experiment

Show

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##          0 5040  388
##          1   24 1029
##
##                Accuracy : 0.9364
##                  95% CI : (0.9302, 0.9422)
##     No Information Rate : 0.7814
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.795
##
##  Mcnemar's Test P-Value : < 2.2e-16
##
##             Sensitivity : 0.7262
##             Specificity : 0.9953
##          Pos Pred Value : 0.9772
##          Neg Pred Value : 0.9285
##              Prevalence : 0.2186
##          Detection Rate : 0.1588
##    Detection Prevalence : 0.1625
##       Balanced Accuracy : 0.8607
##
##        'Positive' Class : 1
##
```

Show

```
##
## Class-Weighted Random Forest Metrics
```

Show

```
## Accuracy : 0.9364
```

Show

```
## Precision: 0.9772
```

Show

```
## Recall   : 0.7262
```

Show

```
## F1-score : 0.8332
```

8.3 Small category counts

Some categories are small, so their default rates are less reliable.

Show

```
##
## Loan Grade Counts:
```

Show

```
##
##     A     B     C     D     E     F     G
## 10702 10384  6436  3619   963   241    64
```

Show

```
##
## Home Ownership Counts:
```

Show

```
##
## MORTGAGE    OTHER      OWN     RENT
##    13366      106     2563    16374
```

Show

```
##
## Default Rates by Loan Grade (%):
```

Show

```
##     A     B     C     D     E     F     G
##  9.96 16.32 20.76 59.05 64.49 70.54 98.44
```

Show

```
##
## Default Rates by Home Ownership (%):
```

Show

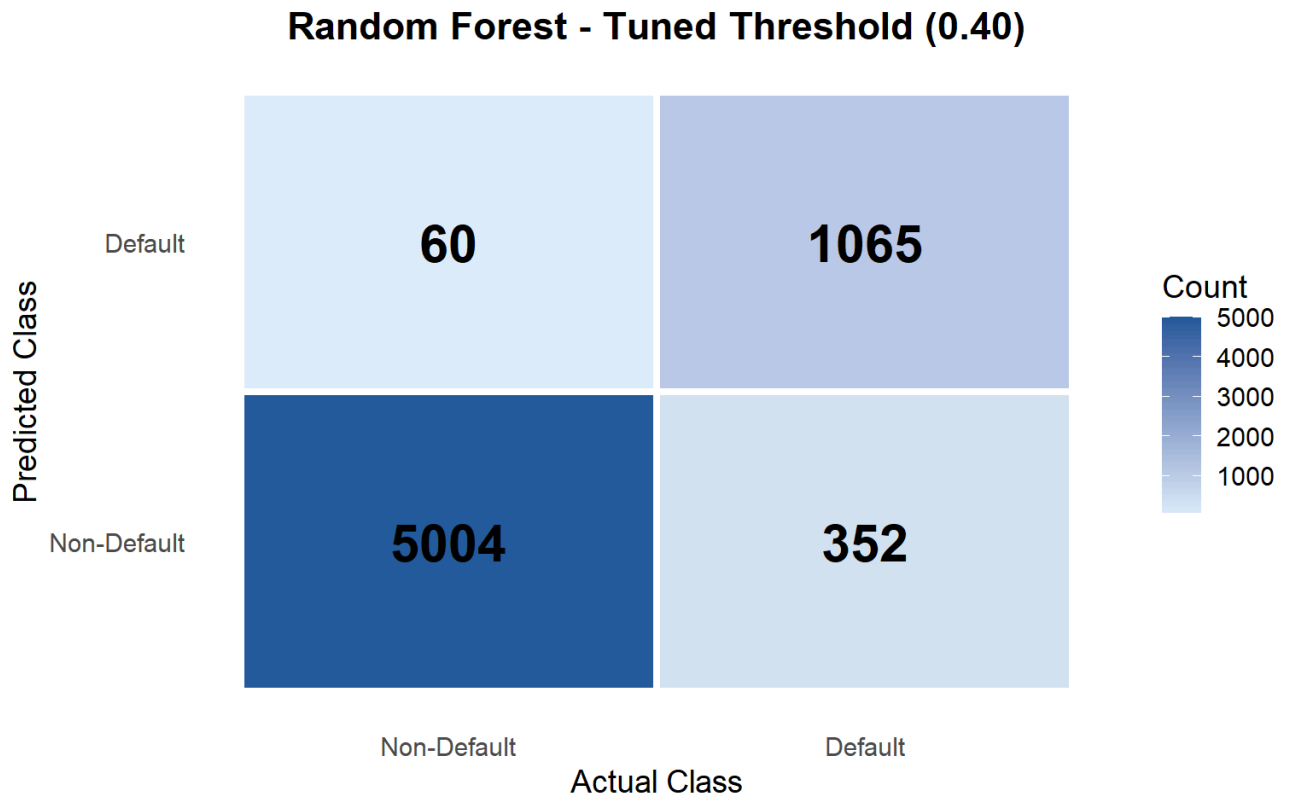
```
## MORTGAGE    OTHER      OWN     RENT
##    12.62    31.13     7.49    31.61
```

8.4 Random Forest evaluation at threshold 0.40

Show

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##          0 5004  352
##          1   60 1065
##
##                Accuracy : 0.9364
##                  95% CI : (0.9302, 0.9422)
##     No Information Rate : 0.7814
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.799
##
##  Mcnemar's Test P-Value : < 2.2e-16
##
##             Sensitivity : 0.7516
##             Specificity : 0.9882
##          Pos Pred Value : 0.9467
##          Neg Pred Value : 0.9343
##              Prevalence : 0.2186
##          Detection Rate : 0.1643
##    Detection Prevalence : 0.1736
##       Balanced Accuracy : 0.8699
##
##        'Positive' Class : 1
##
```

Show



Show

```
##
## Threshold: 0.4
```

Show

```
## Accuracy         : 0.9364
```

Show

```
## Precision        : 0.9467
```

Show

```
## Recall           : 0.7516
```

Show

```
## F1-score         : 0.8379
```

Show

```
## Balanced accuracy: 0.8699
```

8.5 Reported results vs results from this knit

The “Reported” columns are the figures from the original project run. The “This knit” columns are computed live above. Small differences can occur with different R or package versions.

Show

```
##                Model Reported_Accuracy Knit_Accuracy Reported_Precision
##   Logistic Regression            0.8619        0.8619             0.7554
##  Random Forest (0.50)            0.9375        0.9375             0.9765
##  Random Forest (0.40)            0.9364        0.9364             0.9467
##  Knit_Precision Reported_Recall Knit_Recall Reported_F1 Knit_F1 Reported_AUC
##          0.7554          0.5448      0.5448      0.6330  0.6330       0.8639
##          0.9765          0.7318      0.7318      0.8366  0.8366       0.9384
##          0.9467          0.7516      0.7516      0.8379  0.8379           NA
##  Knit_AUC
##    0.8639
##    0.9384
##        NA
```

9 LIME Explainability

LIME fits a simple local model around one test applicant to show which feature ranges pushed the Random Forest towards “default” for that case. Two small S3 methods tell `lime` how to treat a `randomForest` model and how to obtain class probabilities (columns named `"0"` and `"1"`). A seed is set so the explanation can be repeated; LIME samples randomly, so the numbers can still differ slightly between package versions.

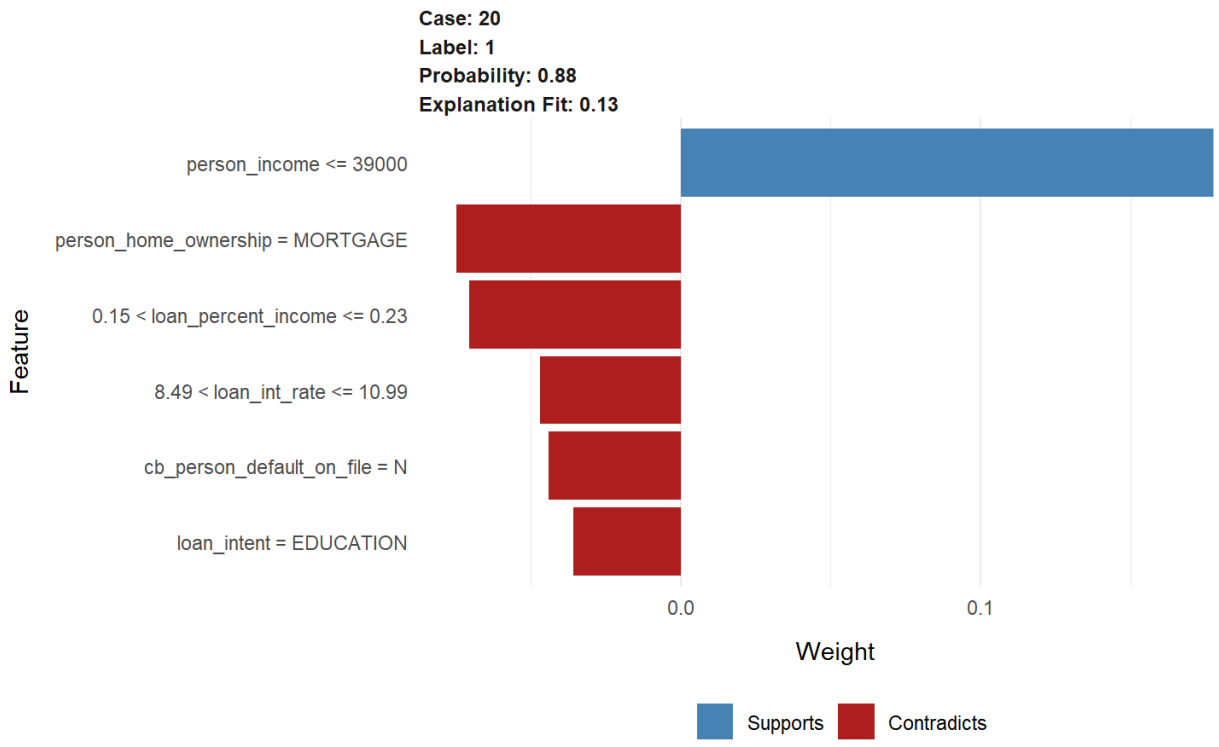
Show

```
##       0    1
## 20 0.12 0.88
```

Show

```
##                    feature feature_value                      feature_desc
## 1            person_income      10800.00            person_income <= 39000
## 2     person_home_ownership          1.00   person_home_ownership = MORTGAGE
## 3       loan_percent_income          0.16 0.15 < loan_percent_income <= 0.23
## 4            loan_int_rate         10.99      8.49 < loan_int_rate <= 10.99
## 5 cb_person_default_on_file          1.00      cb_person_default_on_file = N
## 6               loan_intent          2.00            loan_intent = EDUCATION
##   feature_weight label_prob  model_r2
## 1     0.17780245       0.88 0.1277161
## 2    -0.07466729       0.88 0.1277161
## 3    -0.07039499       0.88 0.1277161
## 4    -0.04697310       0.88 0.1277161
## 5    -0.04391272       0.88 0.1277161
## 6    -0.03575477       0.88 0.1277161
```

Show



Show

```
## Applicant index: 2
```

Show

```
## Predicted class: 1
```

Show

```
## Actual class: 1
```

The original run reported a model probability of about 0.88 for this case and a LIME explanation fit (`model_r2`) of about 0.14, which is low. A low fit means the local linear model explains little of the forest's behaviour around this applicant, so the individual feature contributions should be read cautiously.

10 Conclusion

Data preparation. The original file has 32,581 rows and 12 variables. After removing 165 duplicate rows and a few invalid records (ages above 100 and employment lengths above 50 years), 32,409 rows remain. Default cases make up 7,088 rows (21.87%), so the classes are imbalanced. A stratified 80:20 split gives 25,928 training and 6,481 test records, and missing values in the models are filled with training-set medians.

Exploratory and statistical findings. Default rates differ across loan grades, loan purposes, home-ownership groups and previous-default history, and the chi-square tests found a statistically significant association between each categorical variable and loan status. These are associations, not causal relationships.

Baseline model comparison (reported results).

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	86.19%	75.54%	54.48%	0.6330	0.8639
Random Forest (threshold 0.50)	93.75%	97.65%	73.18%	0.8366	0.9384

Random Forest outperformed Logistic Regression on every reported metric.

Threshold-tuning experiment. At a threshold of 0.40 the Random Forest reported 93.64% accuracy, 94.67% precision, 75.16% recall, an F1-score of 0.8379 and 86.99% balanced accuracy (confusion matrix: TN = 5,004, FP = 60, FN = 352, TP = 1,065). Compared with 0.50, it catches more defaults at the cost of more false positives. The 0.40 threshold was selected after examining test-set results, so this is an experiment rather than an unbiased final estimate.

Class-weighted Random Forest. Giving the default class a weight of 2 did not improve recall in the reported run (about 72.6% versus 73.2% at the baseline).

LIME. LIME was applied to one test applicant who was predicted as default and was an actual default. Its explanation fit was low, so the explanation is only indicative.

10.1 Limitations and future work

- Default cases are about 21.87% of the cleaned data, so class imbalance remains relevant.
- Loan grade G contains 64 records and the `OTHER` home-ownership group contains 106 records; findings for these small groups need caution.
- Threshold selection used the test set; it should be done on a validation set in future work.
- Associations, feature importance and LIME explanations do not establish causal root causes.
- The LIME example had a low explanation fit.
- Results depend on this dataset and one random split, and may not generalise to other lending populations or time periods.
- Future work: validation-based threshold selection, cross-validation, other imbalance-handling methods, and testing on new data.

11 Dataset Source and Project Repository

- **Dataset:** Credit Risk Dataset (`credit_risk_dataset.csv`) - https://www.kaggle.com/datasets/laotse/credit-risk-dataset (https://www.kaggle.com/datasets/laotse/credit-risk-dataset)
- **GitHub repository:** https://github.com/Gautam-2107/Loan-Default-Risk-Analysis (https://github.com/Gautam-2107/Loan-Default-Risk-Analysis)

12 Reproducibility Instructions

1. Download or clone the project folder so that `Loan_Default_Risk_Analysis_Final.Rmd` and `credit_risk_dataset.csv` are in the **same folder**.
2. Open `Loan_Default_Risk_Analysis_Final.Rmd` in RStudio.
3. Make sure these R packages are installed (the first chunk installs any that are missing, which needs an internet connection): `dplyr` , `ggplot2` , `corrplot` , `caret` , `pROC` , `randomForest` , `lime` .
4. Click **Knit > Knit to HTML**. The output `Loan_Default_Risk_Analysis_Final.html` is created next to the `.Rmd` file.
5. Random seeds are fixed (`set.seed(123)`) before the train-test split, each Random Forest and the LIME explanation, so a re-run with the same R and package versions should reproduce the same results. Training the two Random Forests can take a few minutes.